

## PRIMARY 3 BASIC SCIENCE SCHEME OF WORK

### FIRST TERM

| Week | Topic                      | Content                                                                                                                                                  |
|------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Measurement of Length I    | Introduction to measurement; Meaning of length; Importance of measuring length; Traditional units of measuring length (handspan, footstep, arm's length) |
| 2    | Measurement of Length II   | Standard units of measuring length (centimeter, meter); Instruments for measuring length (ruler, tape measure, meter rule); Practical activities         |
| 3    | Measurement of Mass I      | Meaning of mass; Difference between mass and weight; Importance of measuring mass; Traditional ways of measuring mass (calabash, cup)                    |
| 4    | Measurement of Mass II     | Standard units of measuring mass (gram, kilogram); Instruments for measuring mass (scale, balance, weighing scale); Practical activities                 |
| 5    | <b>Midterm Examination</b> | Assessment of weeks 1-4 topics                                                                                                                           |
| 6    | <b>Midterm Break</b>       |                                                                                                                                                          |
| 7    | Measurement of Time I      | Meaning of time; Importance of measuring time; Traditional ways of telling time (cock crow, sun position, shadows)                                       |
| 8    | Measurement of Time II     | Modern instruments for measuring time (clock, watch, timer, calendar); Units of time (second, minute, hour, day, week, month, year)                      |
| 9    | Measurement of Time III    | Reading the clock; Telling time (o'clock, half past); Days of the week; Months of the year; Seasons                                                      |
| 10   | Soil I                     | Meaning of soil; Importance of soil; Where soil comes from; Different colors of soil (red, black, brown, grey)                                           |
| 11   | <b>Revision</b>            | Review of all topics covered in first term                                                                                                               |
| 12   | <b>Examination</b>         | End of first term assessment                                                                                                                             |
| 13   | <b>Closing</b>             | Collection of results; Distribution of report cards; Closing                                                                                             |

|  |  |          |
|--|--|----------|
|  |  | ceremony |
|--|--|----------|



## PRIMARY 3 BASIC SCIENCE LESSON NOTES

### FIRST TERM

---

#### WEEK 1: MEASUREMENT OF LENGTH I.

**OBJECTIVES** By the end of the lesson, pupils should be able to:

- Define the term "measurement."
- Explain the meaning of length as the distance between two points.
- State two reasons why we measure length.
- Identify and demonstrate three traditional methods of measuring length.

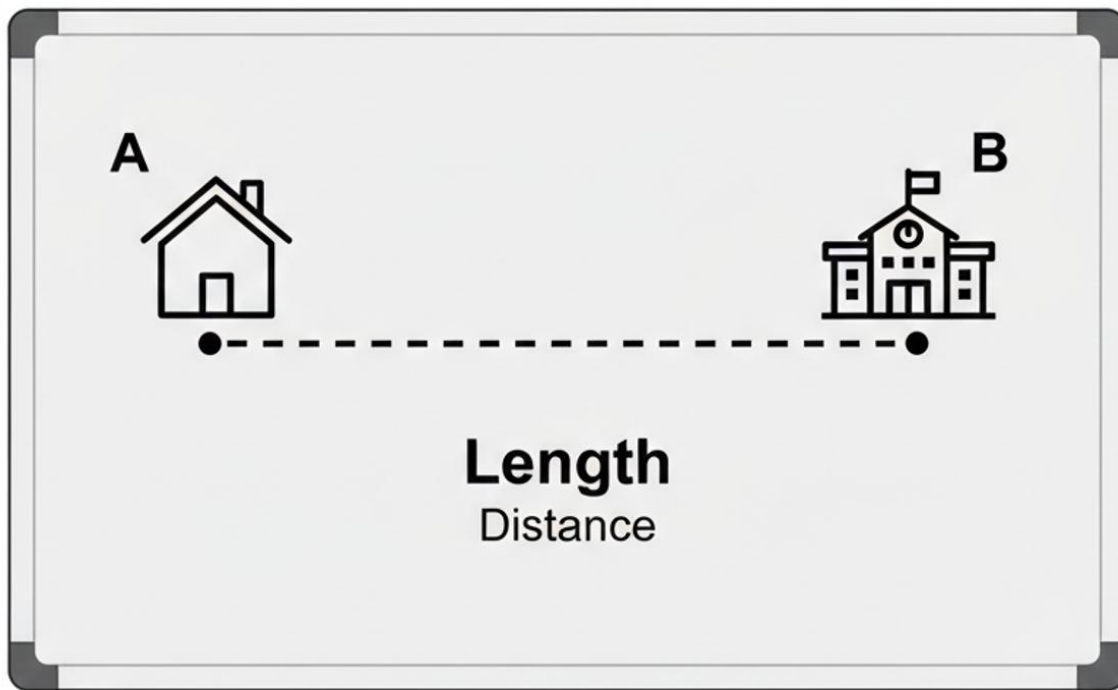
#### CONTENT

**Introduction to Measurement:** Measurement means finding out the size, amount, or degree of something. It is like asking questions: "How long is it?" "How heavy is it?" or "How much is inside?"

- We measure things every day to know their size so we can use them correctly.
- For example, before you buy a new school bag, you want to know if all your books will fit inside. That is measurement.

**Meaning of Length:** Length tells us how long or short an object is. It is the distance from one point to another. Think about your classroom.

- The distance from the door to the window is a length.
- The distance from your desk to your friend's desk is also a length. Length helps us compare things.
- A pencil is **shorter** than a ruler. A football field is **longer** than our classroom.

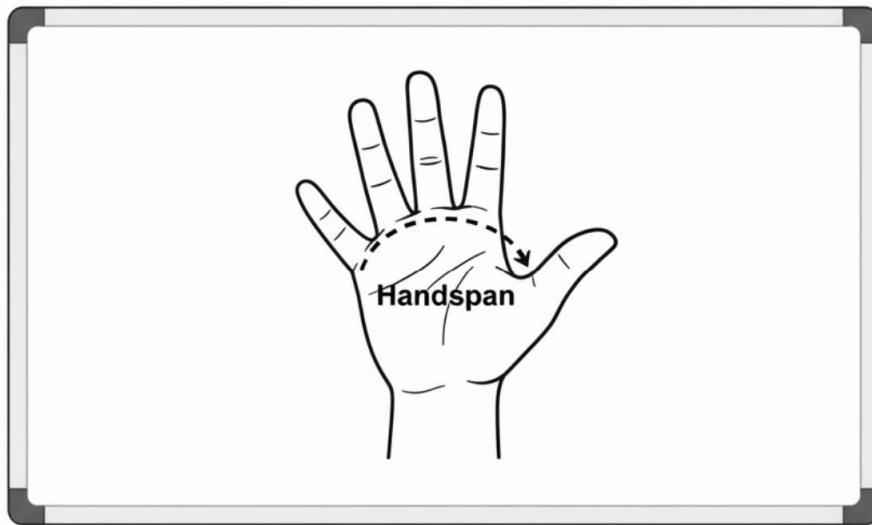


**Importance of Measuring Length:** We measure length for many important reasons in our daily lives:

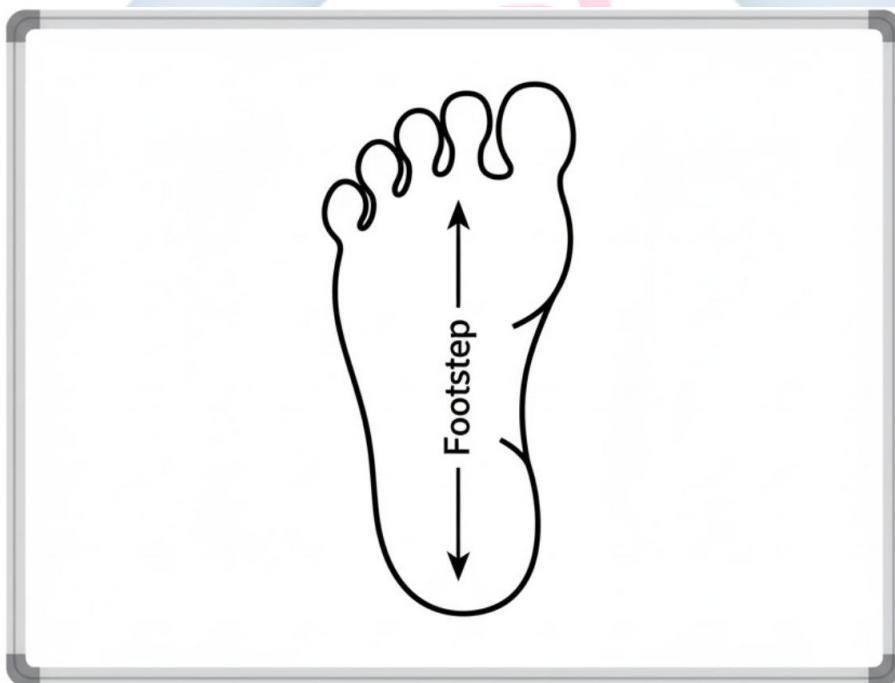
- **For Clothes:** A tailor measures the length of your arms and legs to make a school uniform that fits you perfectly.
- **For Building:** Builders measure the length and width of land to build strong houses, classrooms, and fences that are straight and the right size.
- **For Furniture:** Your parents measure the length of a room to know if a new bed or cupboard will fit in the space.
- **For Travel:** We measure the distance from home to school to know how far we walk or drive every day.

**Traditional Units of Measurement:** Long ago, before we had rulers and tape measures, people used parts of their own bodies to measure things. These are called **non-standard units** because they are not the same for everyone. A child's hand is smaller than an adult's hand, so their measurements will be different.

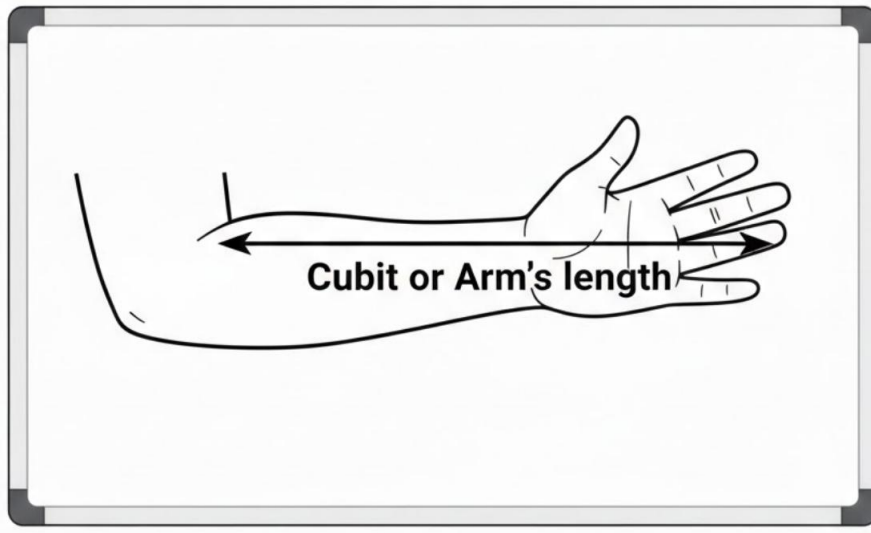
- **Handspan:** Stretch your hand wide. The distance from the tip of your thumb to the tip of your little finger is your handspan.



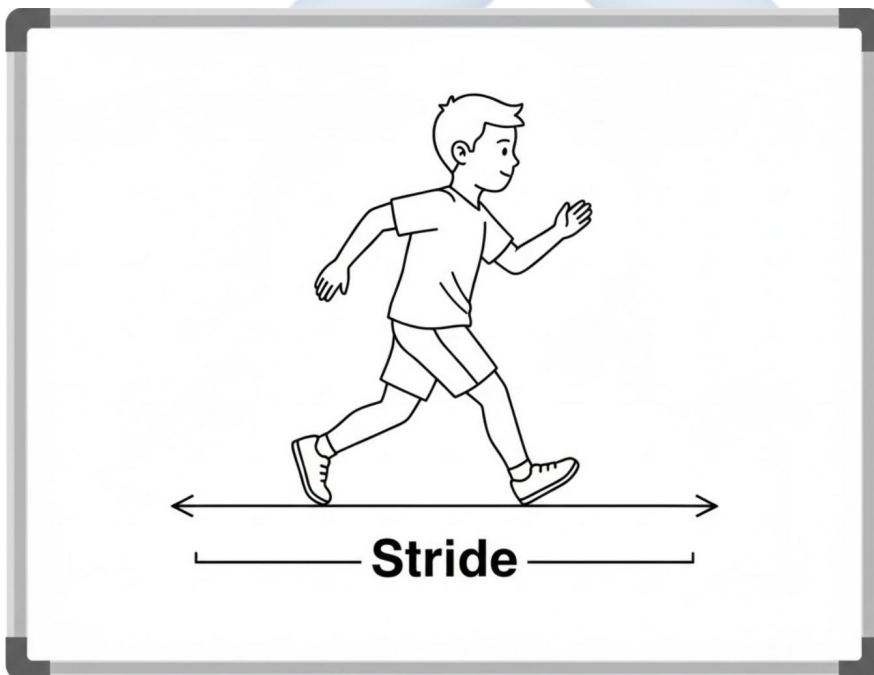
- **Footstep:** This is the length of your foot, or the length of one step you take while walking normally.



- **Arm's Length (Cubit):** Bend your arm. The distance from your elbow to the tip of your middle finger is one cubit.



- **Stride:** Take a big, long step. The distance from where your back foot leaves the ground to where your front foot lands is your stride.



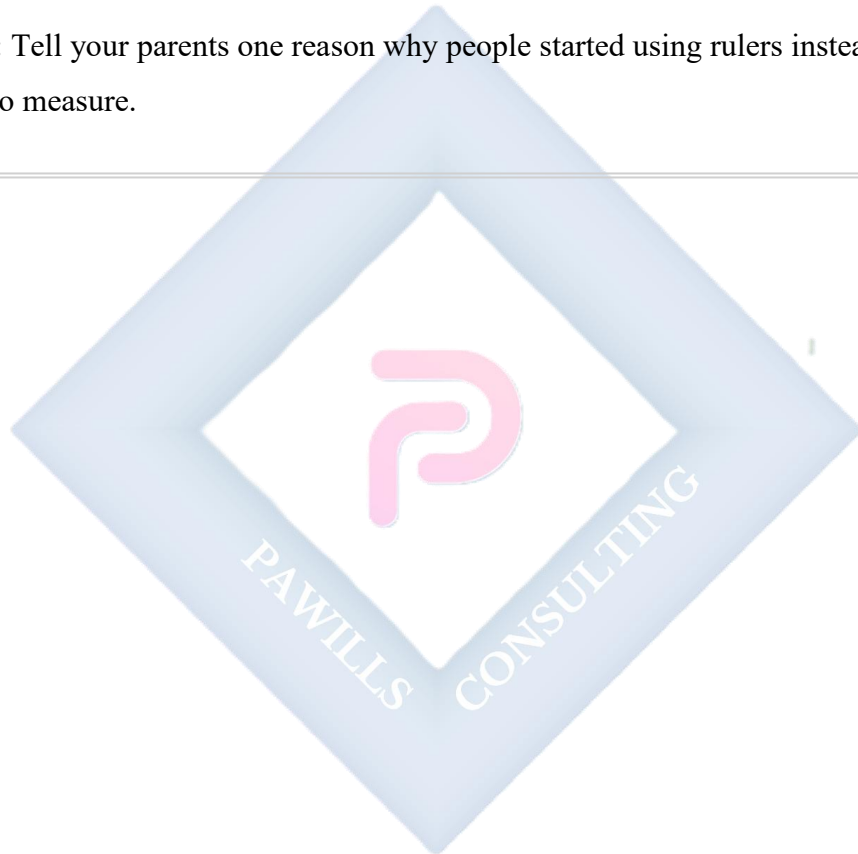
## EVALUATION

1. In your own words, what is measurement?
2. If you want to know how far it is from your desk to the teacher's table, are you trying to find its length or its weight?
3. Who needs to measure length to make your school uniform?

4. Name two traditional ways of measuring length that use body parts.
5. Why is a handspan not a good way for a builder to measure a wall? (Hint: Think about different people measuring the same wall).

### ASSIGNMENT

- **(Practical):** Use your handspan to measure the length of your reading table at home. Write down the number of handspans.
- **(Drawing):** Draw a simple picture of a hand showing what a handspan is.
- **(Oral):** Tell your parents one reason why people started using rulers instead of just their hands to measure.



## WEEK 2: MEASUREMENT OF LENGTH II.

**OBJECTIVES** By the end of the lesson, pupils should be able to:

- Identify standard units for measuring length (centimeter and meter).
- List three modern instruments used for measuring length.
- Measure classroom objects using a ruler or tape measure.
- Differentiate between the uses of a ruler and a tape measure.

### CONTENT

**Standard Units of Measurement:** Because our body parts are different sizes, traditional measurements can be confusing. If I say a table is 10 handspans long, it might be true for me but not for you! To solve this, people all over the world agreed to use the same, fixed units. These are called **standard units**.

- **Centimeter (cm):** This is a very small unit. We use it to measure short things. Look at your ruler; the small lines between the numbers are centimeters.
- **Examples:** A pencil, a crayon, an eraser, your finger.
- **Meter (m):** This is a bigger unit. One meter is equal to 100 centimeters. We use it to measure longer things.
- **Examples:** The length of the classroom, your height, a long piece of cloth.



## MEASUREMENT OF LENGTH

### SMALLER OBJECTS



Measured in cm  
e.g., 15 cm

### TALLER OBJECTS

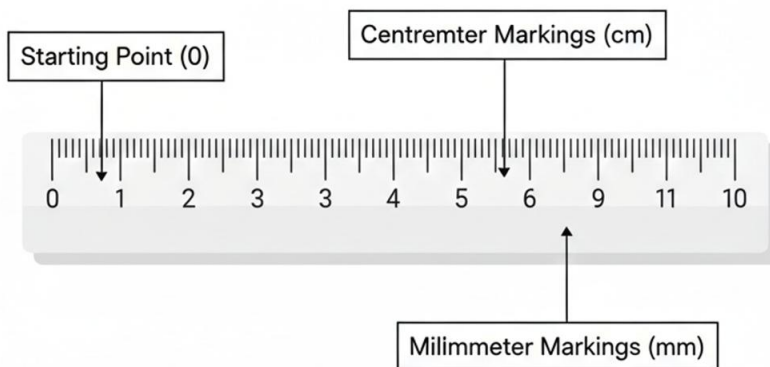


Measured in m  
e.g., 12 m

**Instruments for Measuring Length:** We use special tools to measure length accurately with these standard units.

- **Ruler:** A flat, straight stick made of wood or plastic. It is usually 30 cm long. We use it on our desks to draw straight lines and measure small objects like books and pencils.

## PARTS A RULER



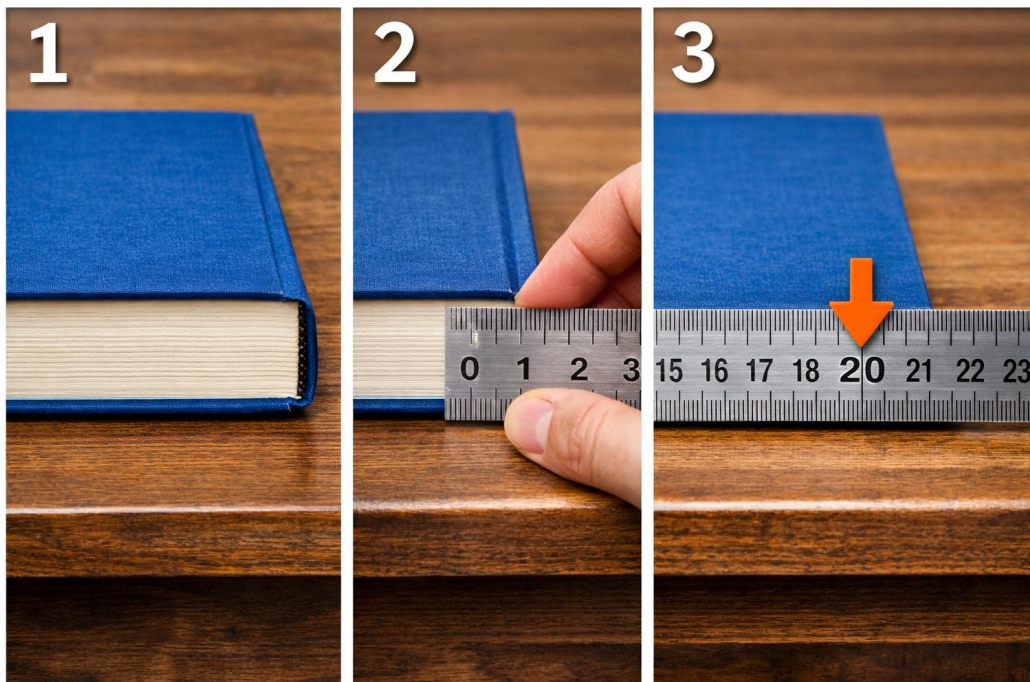
- **Tape Measure:** A long, thin, flexible strip of metal or cloth that can roll up. It can measure curved surfaces. Tailors use it to measure around your waist, arms, and chest to make clothes.



- **Meter Rule:** A long, straight wooden or plastic stick that is exactly one meter long. Teachers use it on the chalkboard. Carpenters use it to measure wood.
- **Carpenter's Tape/Retractable Tape:** A metal tape inside a small case. You pull the tape out to measure, and it springs back in. It's used for building and fixing things.

### **Practical Activity: How to Measure Correctly**

1. **Step 1:** Place the object you want to measure (e.g., a book) flat on the table.
2. **Step 2:** Place the "0" mark of your ruler at one end of the book.
3. **Step 3:** Look straight down at the ruler. Find the number on the ruler that is at the other end of the book.
4. **Step 4:** That number is the length in centimeters (cm).



## EVALUATION

1. Would you use centimeters (cm) or meters (m) to measure the length of your toothbrush?
2. Which unit (cm or m) is better for telling your friend how tall you are?
3. If a tailor wants to make you a shirt, which measuring instrument will they most likely use?
4. What is the main difference between a ruler and a tape measure? (Hint: Can you bend a ruler?)
5. Your textbook is 25cm long. Is this a standard or non-standard measurement? How do you know?

## ASSIGNMENT

- **(Practical):** Use your ruler to measure the length of your mathematics textbook. Write down the answer in cm.
- **(Practical):** Find a door in your house. Using your footsteps, measure how many footsteps wide it is. Write it down.
- **(Written):** Name two things in your home that you would need a tape measure to measure properly (think of things that are not flat).



## WEEK 3: MEASUREMENT OF MASS I.

**OBJECTIVES** By the end of the lesson, pupils should be able to:

- Define mass as the quantity of matter in an object.
- Distinguish between mass and weight simply.
- State why we measure mass.
- Identify traditional ways of measuring mass.

### CONTENT

**Meaning of Mass:** Mass is the amount of "stuff" or **matter** inside an object. It tells us how much material the object is made of.

- A big, full water bottle has more mass (more water, more plastic) than a small, empty water bottle.
- We often say something is "heavy" or "light," but scientists use the word **mass**.

**Simple Difference Between Mass and Weight:** This can be tricky, but let's keep it simple:

- **Mass:** How much **matter** is in you. This stays the same everywhere—on Earth, on the moon, in space. It is like counting how many blocks are in a box.
- **Weight:** How hard gravity is **pulling** on that mass. Gravity on Earth is strong, so you feel heavy. Gravity on the moon is weaker, so you would feel lighter there, even though your mass (the number of blocks) is the same!
- For now, it is okay to use "heavy" and "light" when talking about everyday things.

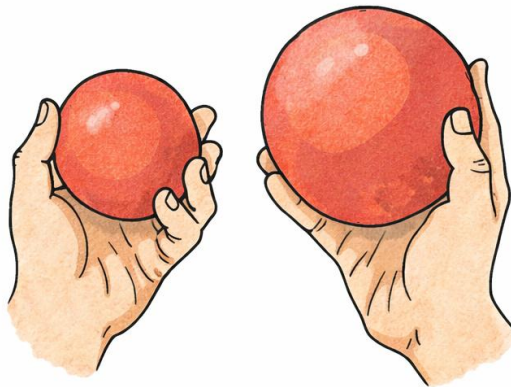
**Importance of Measuring Mass:** We measure mass every day for important reasons:

- **Buying and Selling:** In the market, you buy rice, beans, or meat by their mass (e.g., "Give me 2 kg of rice"). This tells you how much you are getting for your money.
- **Cooking:** Recipes need the right mass of ingredients. Too much or too little salt can spoil the soup!
- **Health:** When you visit the clinic, the nurse measures your mass to see if you are growing well and healthy.



**Traditional Ways of Measuring Mass:** Long ago, people did not have digital scales. They used clever methods:

- **Hefting:** Lifting two objects, one in each hand, to feel which one is heavier. Like comparing a mango and an orange.
- **Using Containers:** They used specific cups, bowls, or calabashes like a "**Mudu**" or "**Congo**" to measure grains like garri, rice, or beans. Everyone in the community knew how much a "Mudu" was.
- **Counter-balances:** Using a simple stick (like a see-saw) with a pan on each end. They would put the item (e.g., yam) on one side and add stones to the other side until they balanced.



Hefting



Using a container

## EVALUATION

1. What does the mass of an object tell us?
2. If you have a big stone and a small stone, which one has more mass?
3. Name one traditional container used in the market to measure food items.

4. Why must a baker measure the mass of flour when making bread?
5. Would you use hefting to compare the mass of a football and a feather? Why or why not?

### ASSIGNMENT

- **(Practical/Drawing):** Go to the kitchen and find a cup or bowl used for measuring food (like a soup cup). Draw it in your notebook.
  - **(Oral):** Ask your parents or grandparents: "What is a 'Mudu' or 'Derica' used for?" Write down their answer.
  - **(Written):** List three items your family buys in the market where the seller measures the quantity (like beans, rice, or garri).
- 



## WEEK 4: MEASUREMENT OF MASS II.

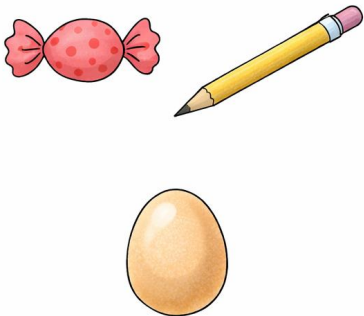
**OBJECTIVES** By the end of the lesson, pupils should be able to:

- Identify standard units of mass (gram and kilogram).
- List instruments used for measuring mass.
- Measure the mass of simple objects using a scale.
- Compare the mass of different objects (heavy vs. light).

### CONTENT

**Standard Units of Mass:** Just like with length, we use standard units for mass so everyone gets the same amount.

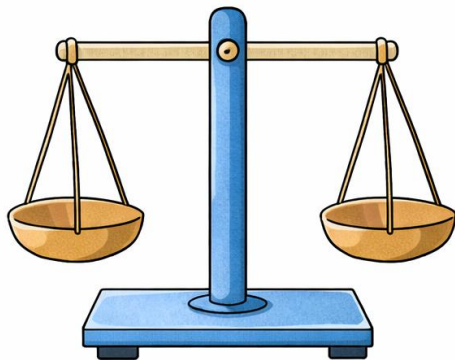
- **Gram (g):** A very light unit. Used for small, light things.
- **Examples:** A sachet of sugar (10g), a pencil, a key, a slice of bread.
- **Kilogram (kg):** A heavier unit. 1 kilogram = 1000 grams. Used for heavier things.
- **Examples:** A bag of rice (5kg, 10kg), a bunch of plantains, a newborn baby.

| Gram (g)                                                                            | Kilogram (kg)                                                                        |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
|  |  |

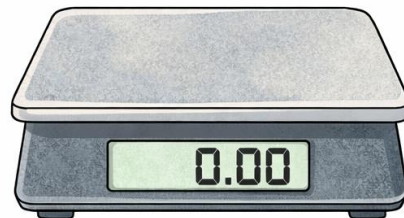


**Instruments for Measuring Mass (Scales):** We use different types of **scales** to find mass accurately.

- **Kitchen Scale:** A small scale used to weigh food ingredients like flour, sugar, or meat for cooking.
- **Bathroom Scale:** A scale you stand on to measure your body mass.
- **Spring Balance:** Has a hook and a spring. You hang an object (like a fish or a bag) from the hook, and it shows the mass on a dial.
- **Beam Balance (Old-fashioned scale):** Has two pans. You put the object on one pan and add standard weights to the other pan until they balance.
- **Electronic/Digital Scale:** Shows the mass as numbers on a screen. Very easy to read. Used in modern shops and clinics.



Beam balance



Digital scale

**Practical Activity: Understanding Mass** Let's think about this: Which is heavier, 1 kg of stones or 1 kg of cotton wool?

- **Answer:** They have the **SAME** mass (1 kg)!

- The cotton wool just takes up much more space (it is less dense). This shows that mass is not about size, but about the amount of matter.

## EVALUATION

1. What unit (g or kg) would you use to tell the mass of a bag of cement?
2. What unit (g or kg) would you use to tell the mass of a chocolate bar?
3. Which instrument would a nurse use to check a baby's mass?
4. If you go to the market to buy beef, which instrument will the butcher use?
5. A big, fluffy pillow and a small, hard stone feel different. Can they have the same mass?  
Explain simply.

## ASSIGNMENT

- **(Practical):** Look at the wrapper of any snack (biscuit, sweets) or a bag of sugar at home. Find where it says the mass (e.g., "Net Wt: 50g" or "1kg"). Write it down.
  - **(Drawing):** Draw a simple picture of a weighing scale.
  - **(Oral):** If you wanted to know if you have grown, which scale at home would you use? Describe it to your teacher tomorrow.
-

## **WEEK 5: MIDTERM EXAMINATION.**

**CONTENT:** Assessment on Weeks 1-4 **EVALUATION:** Written Test **ASSIGNMENT:**  
Review corrections.

---



**WEEK 6: MIDTERM BREAK.**

---



## WEEK 7: MEASUREMENT OF TIME I.

**OBJECTIVES** By the end of the lesson, pupils should be able to:

- Define time.
- Explain the importance of measuring time.
- Identify three traditional ways of telling time.
- Describe how the sun helps in telling time.

### CONTENT

**Meaning of Time:** Time is what we use to measure how long something takes to happen.

- It moves from the **past** (what has happened), through the **present** (what is happening now), to the **future** (what will happen).
- Time helps us answer questions like: "How long is break time?" or "When will school close?"

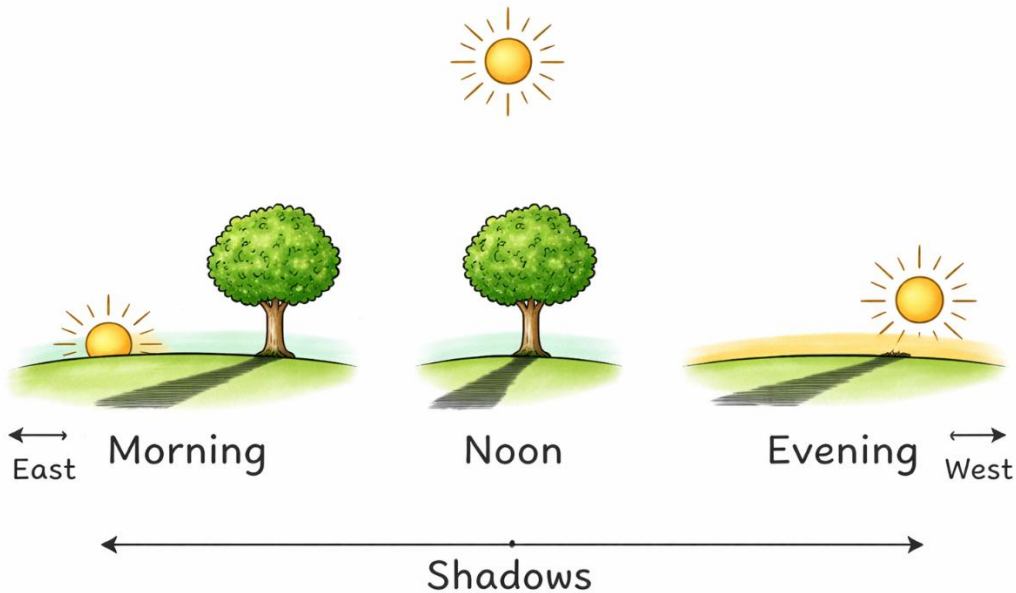
**Importance of Measuring Time:** Knowing the time helps us organize our lives and be responsible:

- **Daily Routine:** To wake up on time for school, to know when it is lunchtime, and when to go to bed.
- **Work and School:** Teachers start and end lessons on time. Offices and markets open and close at specific times.
- **Farming:** Farmers know the right time to plant seeds and the right time to harvest crops.
- **Religion:** It tells us the time for prayers or worship.
- **Appointments:** So we are not late to meet someone.

**Traditional Ways of Telling Time:** Before clocks and watches, people were very clever. They observed nature closely.

- **The Sun (The Most Important):**
- **Morning:** The sun rises in the **East**. Shadows are long and point west.

- **Noon (Midday):** The sun is high in the sky, almost directly above us. Shadows are very short or right under our feet.
- **Evening:** The sun sets in the **West**. Shadows are long again, but now they point east.



- **The Cock Crow:** A rooster (cock) crows loudly very early in the morning, just before sunrise. This was a natural alarm clock for villages.
- **Observing Shadows:** People would stick a tall stick in the ground. By watching how the stick's shadow moved and changed length during the day, they could estimate the time. This is called a **sundial**.
- **Moon Phases:** The changing shape of the moon (from new moon to full moon and back) helped people count about 28-30 days, which is roughly a month.

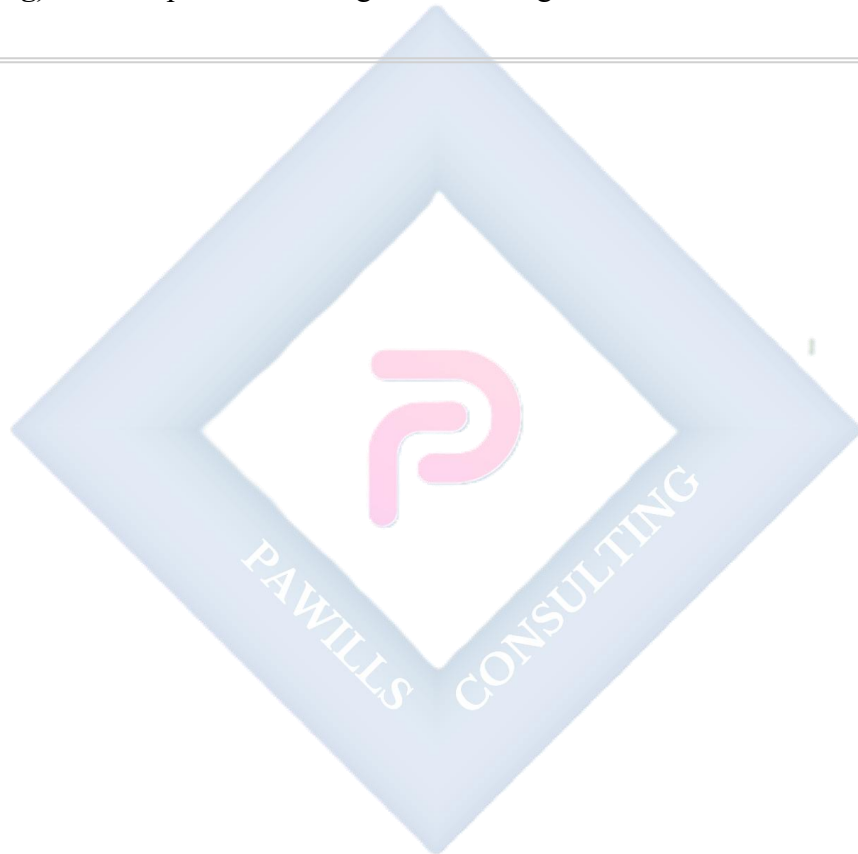
## EVALUATION

1. What does time help us to measure?
2. Give two reasons why it is important for you to know the time.
3. Which bird acted as a natural alarm clock in the olden days?
4. Where is the sun in the sky at noon? What happens to your shadow?

5. Can we use a sundial to tell time on a very cloudy day? Why?

#### ASSIGNMENT

- **(Practical/Observation):** Today, go outside at about 4 pm. Look at your shadow. Is it long or short? Draw a simple picture of yourself and your shadow.
  - **(Oral):** Ask an older person in your family: "How did you know it was time to come home from play when you were little, without a watch?" Share the answer in class.
  - **(Drawing):** Draw a picture showing the sun rising.
- 



## WEEK 8: MEASUREMENT OF TIME II.

**OBJECTIVES** By the end of the lesson, pupils should be able to:

- List four modern instruments for measuring time.
- State the units of measuring time.
- Explain the relationship between seconds, minutes, and hours.
- Identify the use of a calendar.

## CONTENT

**Modern Instruments for Measuring Time:** Now we have accurate machines to tell time anywhere.

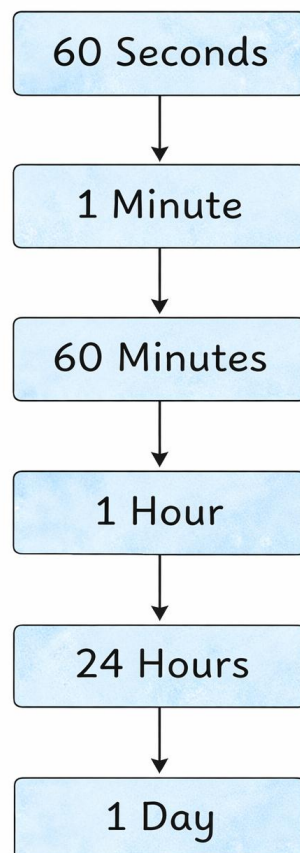
- **Wristwatch:** Worn on the hand. It is portable so you can check time anywhere.
- **Wall Clock:** Hung on the wall in homes, classrooms, and offices for everyone to see.
- **Alarm Clock:** Has a bell or buzzer that rings at a set time to wake you up.
- **Stopwatch:** Used to measure very short periods of time, like in a race. It can start and stop.
- **Mobile Phone:** Most phones show the time and have alarm functions.
- **Calendar:** Shows days, weeks, and months of the year. It helps us plan for future dates.





**Units of Time:** Time is measured in different-sized units, just like we have kobo and naira for money.

- **Second (s):** The basic, smallest unit. It's very short—like saying "one-Mississippi."
- **Minute (min):** 60 seconds make 1 minute.
- **Hour (hr):** 60 minutes make 1 hour.
- **Day:** 24 hours make 1 day (daytime + nighttime).
- **Week:** 7 days make 1 week.
- **Month:** About 28 to 31 days. There are 12 months in a year.
- **Year:** 365 days (or 12 months) make 1 year.



## EVALUATION

1. Which modern time-telling instrument do you wear on your body?
2. What do we use to find out what date Christmas will be next year?
3. How many seconds are there in one minute?
4. How many hours are there from when you wake up to when you go to sleep the next day?  
(Hint: Think about a full day).
5. Which is a longer wait: 30 minutes or 30 seconds?

## ASSIGNMENT

- **(Practical):** Look at a calendar at home. Write down how many days are in this current month.
- **(Drawing):** Draw the face of a clock showing 9 o'clock.
- **(Written):** How many days are in a week? List all seven days in the correct order, starting from Sunday.

## WEEK 9: MEASUREMENT OF TIME III.

**OBJECTIVES** By the end of the lesson, pupils should be able to:

- Read the time on a clock face (Hour and Half Past).
- List the days of the week in order.
- List the months of the year in order.
- Mention the two seasons in Nigeria.

### CONTENT

**Reading the Clock (Analog Clock):** An analog clock has a face with numbers 1 to 12 and two hands.

- **The Short Hand (Hour Hand):** It moves slowly. It tells us the **hour**. When it points exactly to a number, it is that **o'clock**.
- **The Long Hand (Minute Hand):** It moves faster. It tells us the **minutes** past the hour.

**Reading 'O'clock' and 'Half Past':**

- **O'clock:** When the **long (minute) hand points straight up to the number 12**. Look at where the **short (hour) hand** is pointing.
- **Example:** Short hand on 3, long hand on 12 = **3 o'clock**.



- **Half Past:** When the **long (minute) hand points straight down to the number 6**. This means 30 minutes have passed. The hour hand will be **halfway** between two numbers.
- **Example:** Long hand on 6, short hand halfway between 3 and 4 = **Half past 3** (or 3:30).



### Days and Months:

- **Days of the Week (7 days):** Sunday, Monday, Tuesday, Wednesday, Thursday, Friday, Saturday.
- **Months of the Year (12 months):** January, February, March, April, May, June, July, August, September, October, November, December.
- *Tip: Use your knuckles to remember the months with 31 days!*

### Seasons in Nigeria: Nigeria mainly has two seasons:

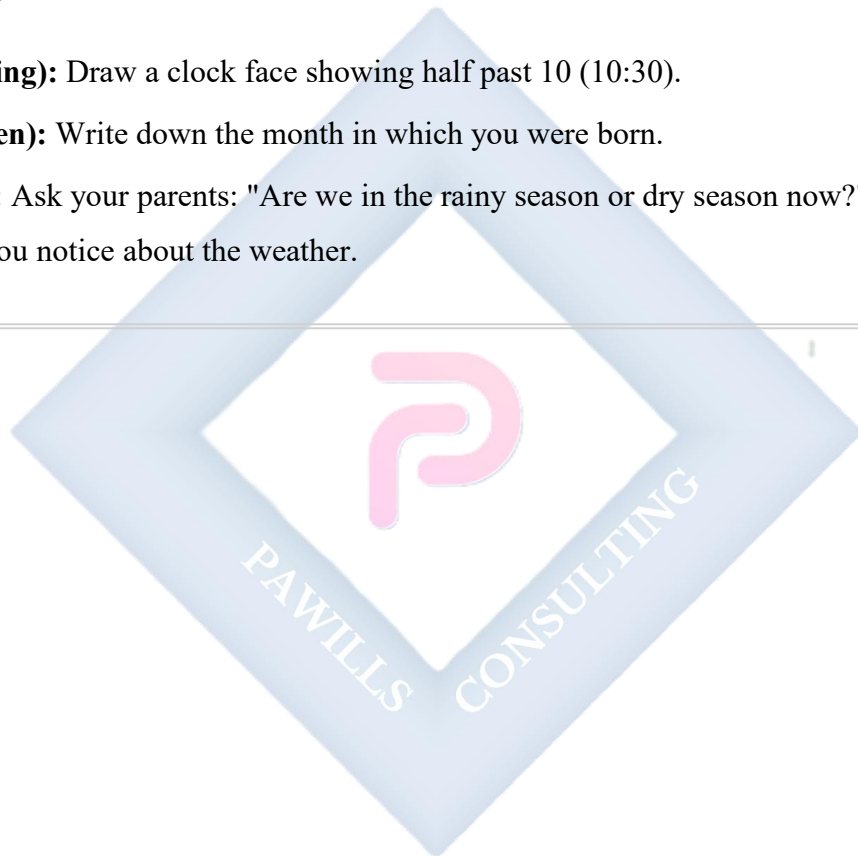
- **Rainy Season:** From around April to October. It rains a lot. The weather is cooler, and farmers plant their crops. Everything looks green.
- **Dry Season:** From around November to March. It is hot and dusty, especially with the **Harmattan** wind from the Sahara Desert. The air is dry, and grasses may turn brown.

## EVALUATION

1. Draw a clock showing 7 o'clock.
2. If the long hand is on 6 and the short hand is between 10 and 11, what time is it?
3. What day comes after Wednesday?
4. Which month is the first month of the year?
5. During which season do we need to carry umbrellas and raincoats?

## ASSIGNMENT

- **(Drawing):** Draw a clock face showing half past 10 (10:30).
  - **(Written):** Write down the month in which you were born.
  - **(Oral):** Ask your parents: "Are we in the rainy season or dry season now?" Explain one thing you notice about the weather.
- 



## WEEK 10: SOIL I.

**OBJECTIVES** By the end of the lesson, pupils should be able to:

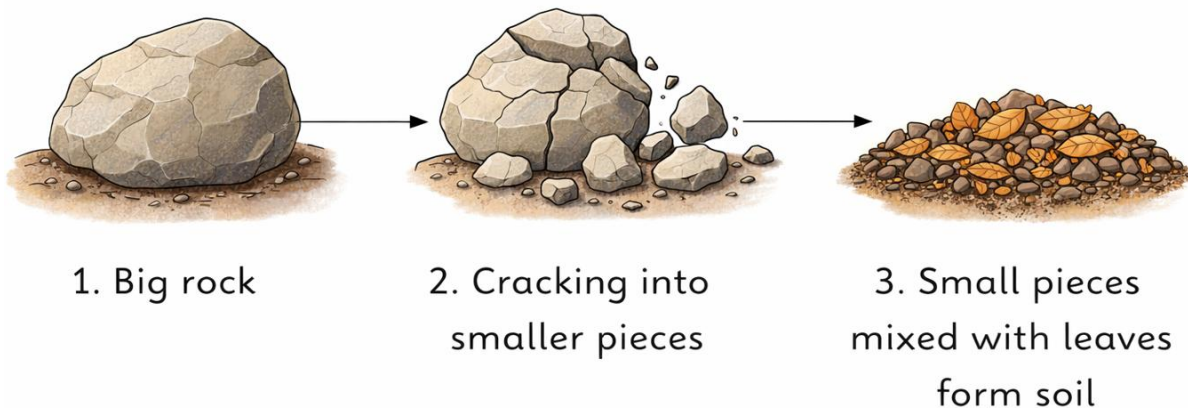
- Define soil.
- State three reasons why soil is important.
- Explain simply where soil comes from.
- Identify different colors of soil.

### CONTENT

**Meaning of Soil:** Soil is the loose, brown (or sometimes black, red, grey) material that covers the land. It is the dirt or ground we walk on, where plants grow, and where some animals live. It is not just dirt; it is a home for many living things!

**Where Does Soil Come From? (Formation):** Soil is formed over a very, very long time. It starts with big, hard **rocks**.

- **Breaking Down:** Sun heats the rocks, rain wets them, wind blows against them, and sometimes water freezes in cracks. Over thousands of years, these forces break big rocks into smaller and smaller pieces.
- **Mixing with Life:** Dead leaves, plants, and animals fall on these tiny rock pieces. They rot and mix with the rock pieces. This rotting material is called **humus**. It makes the soil dark, soft, and good for plants.



**Importance of Soil:** Soil is precious and very useful:

- **For Farming (Food):** All the food we eat comes from plants that grow in soil (like rice, yam) or from animals that eat plants grown in soil.
- **For Building:** We build our houses, schools, and roads on soil. Clay soil is used to make bricks and blocks.
- **Home for Animals:** Many creatures like earthworms, ants, termites, and rabbits live in the soil.
- **For Making Things:** Potters use clay soil to make pots, plates, and cups. Some soils are used to make cement.

**Colors of Soil:** Soil can be different colors because of what is inside it:

- **Dark Brown/Black:** This soil is usually rich in **humus** (decayed plants). It is very fertile and best for growing crops.
- **Red:** Contains a lot of iron. Common in parts of Nigeria (laterite soil). It can be good for building.
- **Grey/Whitish/Yellow:** Often **sandy soil**. Feels gritty. Found near beaches or rivers.
- **Brown/Reddish Grey:** Often **clay soil**. Feels sticky when wet.



**Black**

Good for farming



**Red**

Contains Iron



**Grey**

Sandy



**Brown**

Clay

## EVALUATION

1. What is soil?
2. Complete this sentence: Soil is formed from the breaking down of \_\_\_\_\_ over a long time.
3. Name two things we use soil for.
4. Which color of soil is usually the best for a farmer to grow maize?
5. Name one animal that makes its home in the soil.

## ASSIGNMENT

- **(Practical/Observation):** Go outside with a small container. Collect a little soil from your garden or a nearby patch of land. Look at its color and feel its texture. Write down what you see.
  - **(Drawing):** Draw a plant (like a maize stalk) growing happily in the soil.
  - **(Written):** Complete this important sentence: "Soil gives us \_\_\_\_\_. " (Think of food, house, etc.)
-



**WEEK 11: REVISION.**

**CONTENT:** Review of all Term 1 topics **EVALUATION:** Oral & Written Questions

**ASSIGNMENT:** Study for exams.



## PRIMARY 3 BASIC SCIENCE SCHEME OF WORK

### SECOND TERM

| Week | Topic                      | Content                                                                                                                                  |
|------|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Soil II                    | Types of soil (sandy, clay, loamy); Properties of different types of soil; Texture and color                                             |
| 2    | Soil III                   | Uses of soil (farming, building, pottery); Animals that live in soil; Taking care of soil; Soil erosion and prevention                   |
| 3    | Air in Motion I            | Meaning of air; Air is everywhere; Air occupies space; Air has weight; Simple experiments to show presence of air                        |
| 4    | Air in Motion II           | Wind as air in motion; What causes wind; Types of wind (breeze, storm, hurricane); Uses of wind; Dangers of strong wind                  |
| 5    | <b>Midterm Examination</b> | Assessment of weeks 1-4 topics                                                                                                           |
| 6    | <b>Midterm Break</b>       |                                                                                                                                          |
| 7    | Quality of Water I         | Sources of water (rain, river, stream, well, borehole, tap); Importance of water; Water for drinking, cooking, washing, bathing          |
| 8    | Quality of Water II        | Clean water vs. dirty water; Characteristics of clean water (colorless, odorless, tasteless); Diseases from dirty water                  |
| 9    | Uses of Water              | Domestic uses of water; Water in agriculture; Water in industries; Water for transportation; Water for recreation; Conservation of water |
| 10   | Plants I                   | What are plants? Parts of a plant (root, stem, leaf, flower, fruit, seed); Functions of each plant part                                  |
| 11   | <b>Revision</b>            | Review of all topics covered in second term                                                                                              |
| 12   | <b>Examination</b>         | End of second term assessment                                                                                                            |
| 13   | <b>Closing</b>             | Collection of results; Distribution of report cards; Closing ceremony                                                                    |

## PRIMARY 3 BASIC SCIENCE LESSON NOTES

### SECOND TERM

#### WEEK 1: SOIL II (TYPES AND PROPERTIES).

**OBJECTIVES** By the end of the lesson, pupils should be able to:

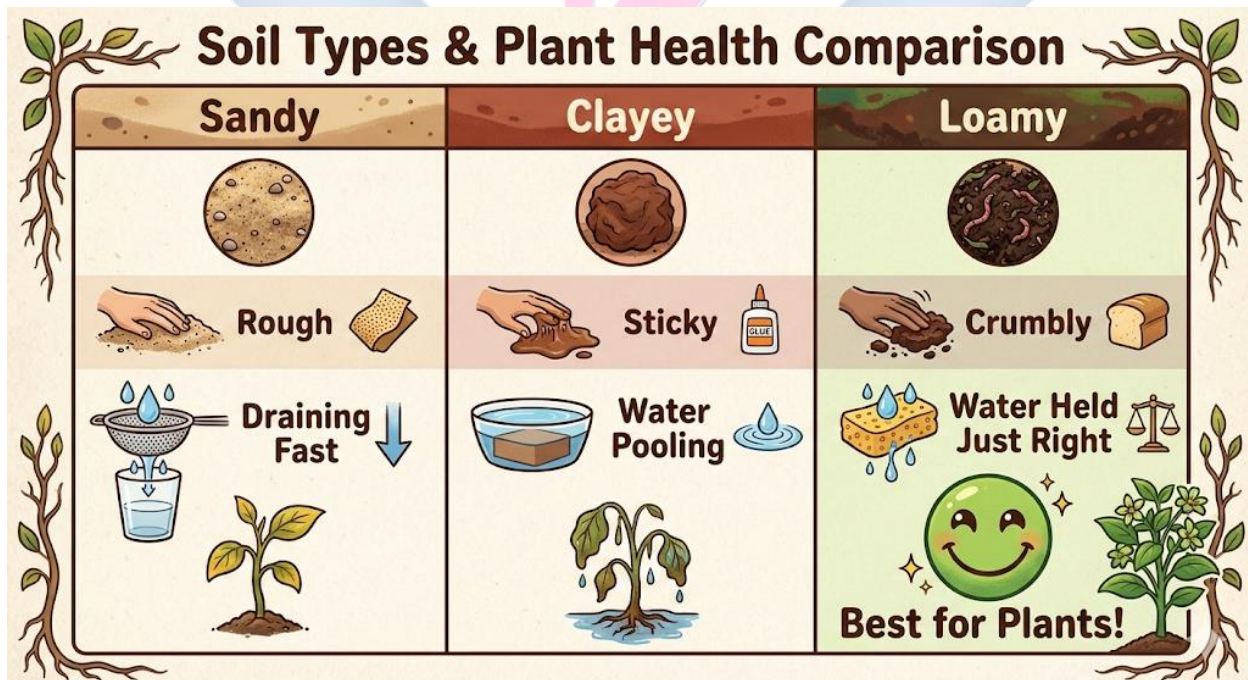
- List the three main types of soil.
- Describe the texture of sandy, clayey, and loamy soil.
- Identify the colour associated with each soil type.
- Differentiate between the soils based on how they hold water.

#### CONTENT

**Types of Soil:** We have different types of soil around us. They feel different, look different, and are used for different things. The three main types are:

- **Sandy Soil:**
- **How it Feels (Texture):** It feels **rough, gritty, and loose**, like the sand at the playground or beach. If you rub it between your fingers, you can feel the tiny grains. The particles are large.
- **Colour:** It is usually **light in colour**—grey, light brown, or yellowish.
- **With Water:** Water passes through sandy soil **very quickly**. If you pour water on it, the water disappears fast into the ground. It does not hold water well. We say it is **very porous**.
- **Where it is Found:** Beaches, deserts, sandpits, and some playgrounds.
- **Clayey Soil:**
- **How it Feels (Texture):** When it is **wet**, it feels **smooth, slippery, and very sticky**—like modeling clay. You can mold it into shapes. When it **dries**, it becomes **very hard and cracks**. The particles are very tiny and packed tightly together.

- **Colour:** It can be **red, brown, or grey**.
- **With Water:** Clayey soil **holds a lot of water** and does not let it pass through easily. It gets waterlogged and heavy. This can drown plant roots.
- **Where it is Found:** Near riverbanks, in swamps, or in areas that get waterlogged.
- **Loamy Soil:**
- **How it Feels (Texture):** This is the **best soil for planting**. It is a perfect **mixture** of sand, clay, and lots of **humus** (decayed plants and animals). It feels **soft, crumbly, and smooth**—like a moist chocolate cake.
- **Colour:** Because of the humus, it is **dark brown or black**.
- **With Water:** It holds **just the right amount of water**—not too much like clay, not too little like sand. It lets extra water drain away so plant roots can breathe and drink.
- **Where it is Found:** In good gardens, farms, and forests.

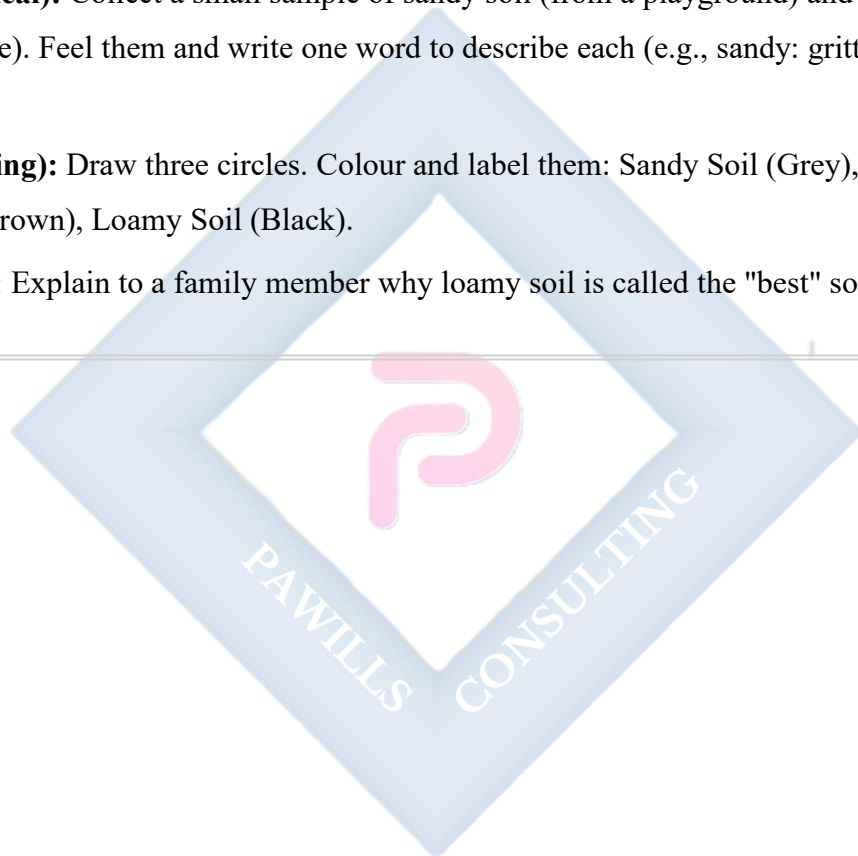


## EVALUATION

1. Which type of soil would stick to your shoes on a rainy day?
2. If you are a farmer, which soil would you choose to plant your vegetables? Why?
3. What gives loamy soil its dark black or brown colour?
4. Why would a plant growing in sandy soil need to be watered very often?
5. Mention the three types of soil.

## ASSIGNMENT

- **(Practical):** Collect a small sample of sandy soil (from a playground) and clayey soil (if possible). Feel them and write one word to describe each (e.g., sandy: gritty, clayey: sticky).
- **(Drawing):** Draw three circles. Colour and label them: Sandy Soil (Grey), Clayey Soil (Red/Brown), Loamy Soil (Black).
- **(Oral):** Explain to a family member why loamy soil is called the "best" soil.



## WEEK 2: SOIL III (USES AND CARE).

**OBJECTIVES** By the end of the lesson, pupils should be able to:

- State three uses of soil.
- Mention two animals that live in the soil.
- Define soil erosion simply.
- List two ways to prevent soil erosion.

## CONTENT

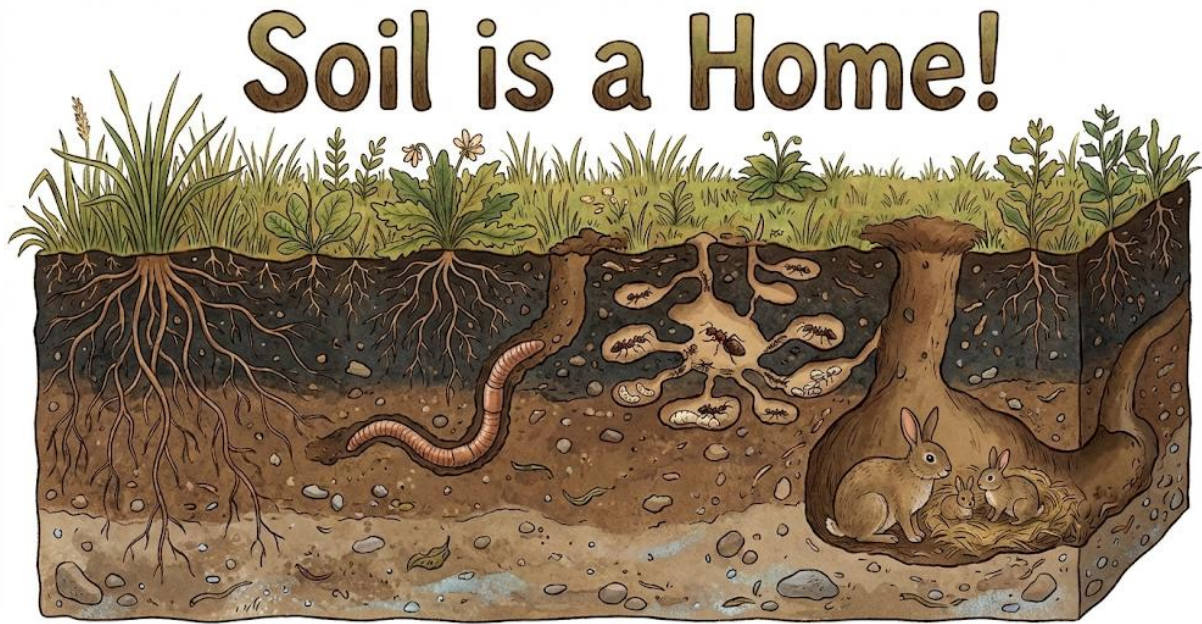
**Uses of Soil:** Soil is not just dirt; it is incredibly useful!

- **For Farming (Food):** This is the most important use. Good loamy soil provides nutrients and a home for plant roots, giving us all our crops—yam, cassava, maize, vegetables, and fruits.
- **For Building:**
- **Laterite Soil (Reddish):** Used to make blocks and bricks for building houses and walls.
- **Sand:** Mixed with cement and gravel to make concrete for building strong foundations, floors, and pillars.
- **Clay:** Used to make bricks and tiles.
- **For Pottery:** Potters use sticky clay soil to mold beautiful pots, plates, cups, and vases. They fire them in a kiln to make them hard.
- **For Making Glass:** Sandy soil (which contains silica) is melted at very high temperatures to make glass for windows, bottles, and mirrors.
- **As a Filter:** Soil can help clean water as it trickles down through the layers to become groundwater.

**Animals in the Soil:** Soil is a home, or habitat, for many animals. These animals are very important for keeping the soil healthy.



- **Earthworms:** They tunnel through the soil, which lets air and water in. Their waste (castings) is excellent fertilizer.
- **Ants and Termites:** They build tunnels and help break down dead leaves.
- **Burrowing Animals:** Rabbits, rats, and moles dig holes (burrows) in the soil to live in.
- **Beetles and Millipedes:** They help to break down dead plants.



**Taking Care of Soil: Preventing Soil Erosion:** We must protect our soil. A big problem is **Soil Erosion**.

- **What is Soil Erosion?** It is the **washing or blowing away** of the top layer of fertile soil by water (rain, rivers) or wind. This leaves the land bare and poor for farming.
- **How to Prevent Soil Erosion (Soil Conservation):**
- **Plant Trees and Grasses:** The roots of plants act like a net, holding the soil tightly together so wind and water cannot carry it away.
- **Build Terraces:** On hilly farms, farmers make flat steps (terraces) to slow down running water.
- **Make Drainage Channels:** Gutters and channels help to direct water flow so it doesn't wash soil away.

- **Avoid Overgrazing:** Too many animals eating all the grass leaves soil bare and open to erosion.

## EVALUATION

1. Which type of soil would a potter use to make a clay pot?
2. Name one animal that lives in the soil and helps to make it better for plants.
3. What do we call it when rain washes away the good topsoil from a farm?
4. How does planting trees help to fight soil erosion?
5. What is sand mixed with to make concrete for building?

## ASSIGNMENT

- **(Written):** Look around your house. List three things you can see that are made from soil or materials that come from soil (e.g., clay pot, glass window, brick wall).
  - **(Drawing):** Draw an earthworm in the soil.
  - **(Oral):** Ask a farmer or gardener in your community: "What do you do to make sure the soil on your farm stays good and doesn't wash away?"
-



### WEEK 3: AIR IN MOTION I.

**OBJECTIVES** By the end of the lesson, pupils should be able to:

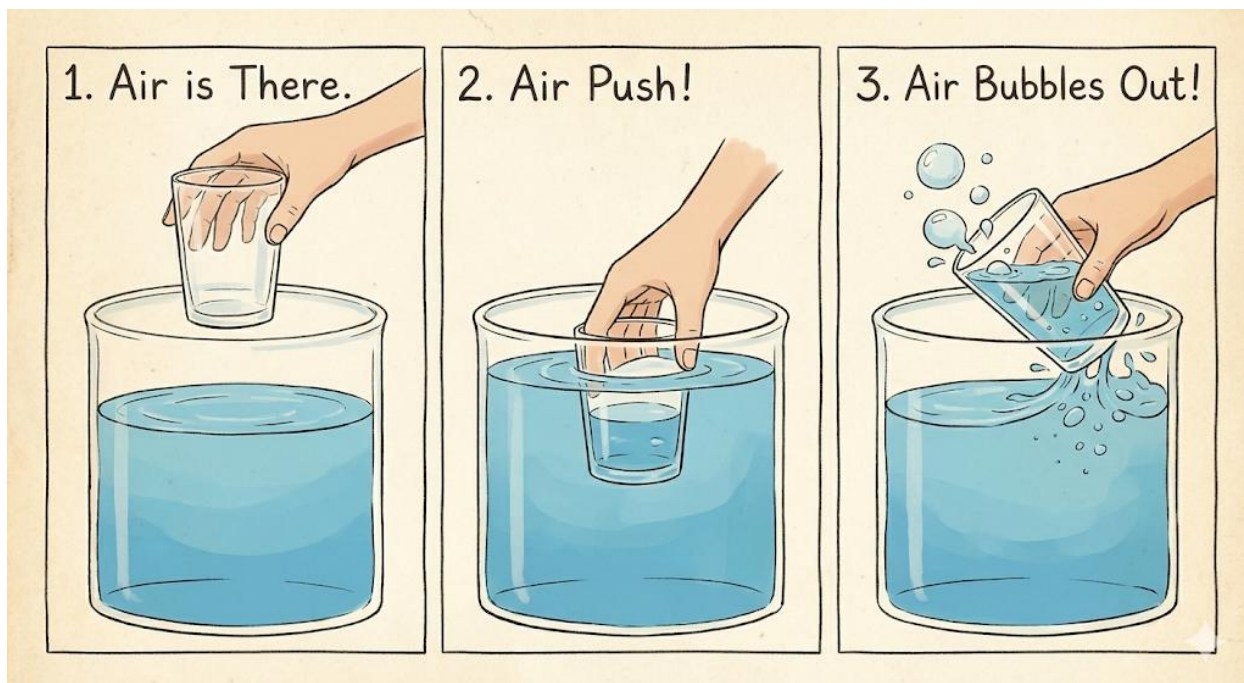
- Define Air.
- State three properties of air (everywhere, occupies space, has weight).
- Perform a simple experiment to show air occupies space.

### CONTENT

**Meaning of Air:** Air is all around us. It is a **mixture of gases** that we cannot see, taste, or usually smell. But we know it is there because we can **feel it** when it moves (wind) and we **breathe it** to stay alive. The most important gases in air for us are **oxygen** (which we breathe in) and **carbon dioxide** (which plants breathe in).

### Properties of Air

- **Air is Everywhere (It is Present All Around):** Air fills every space that is not already filled by something solid or liquid. It is in this classroom, inside an "empty" cup, in water (tiny bubbles), and even in the tiny spaces in soil.
- *Think: An "empty" bottle is actually full of air!*
- **Air Occupies Space (It Takes Up Room):** Even though we can't see it, air takes up space like any other thing.
- **Experiment 1: The Balloon.** Blow air into a balloon. The balloon gets bigger because the air you are blowing is filling up the space inside it.
- **Experiment 2: The Upside-Down Cup.** Push an empty glass cup straight down into a bucket of water. What happens? The water cannot fill the cup because the air inside the cup is occupying the space. Tilt the cup, and you will see bubbles escape as air leaves and water takes its place.



- **Air Has Weight (It is Not Nothing):** Air may feel like nothing, but it has a tiny weight.
- **Demonstration:** Take two identical balloons. Blow one up fully and tie it. Leave the other one flat. Hang them on opposite ends of a ruler or stick (like a seesaw). The side with the blown-up balloon will dip down because it is heavier—it has more air inside, and that air has weight.

## EVALUATION

1. Can we see air? What proof do we have that air exists?
2. When you blow up a balloon, what are you filling it with?
3. In the upside-down cup experiment, why doesn't water rush into the cup immediately?
4. True or False: An empty box is truly empty inside.
5. Where can we find air? (Give two examples).

## ASSIGNMENT

- **(Practical Experiment):** Do the upside-down cup experiment in a bowl of water at home. Observe what happens. Write one sentence about your observation.
- **(Practical):** Use a piece of paper or a fan to wave air towards your face. Write down what you feel.

- **(Drawing):** Draw a picture of a balloon before and after it is blown up.
- 



## WEEK 4: AIR IN MOTION II (WIND).

**OBJECTIVES** By the end of the lesson, pupils should be able to:

- Define Wind.
- List three types of wind.
- State two uses of wind.
- Mention two dangers of strong wind.

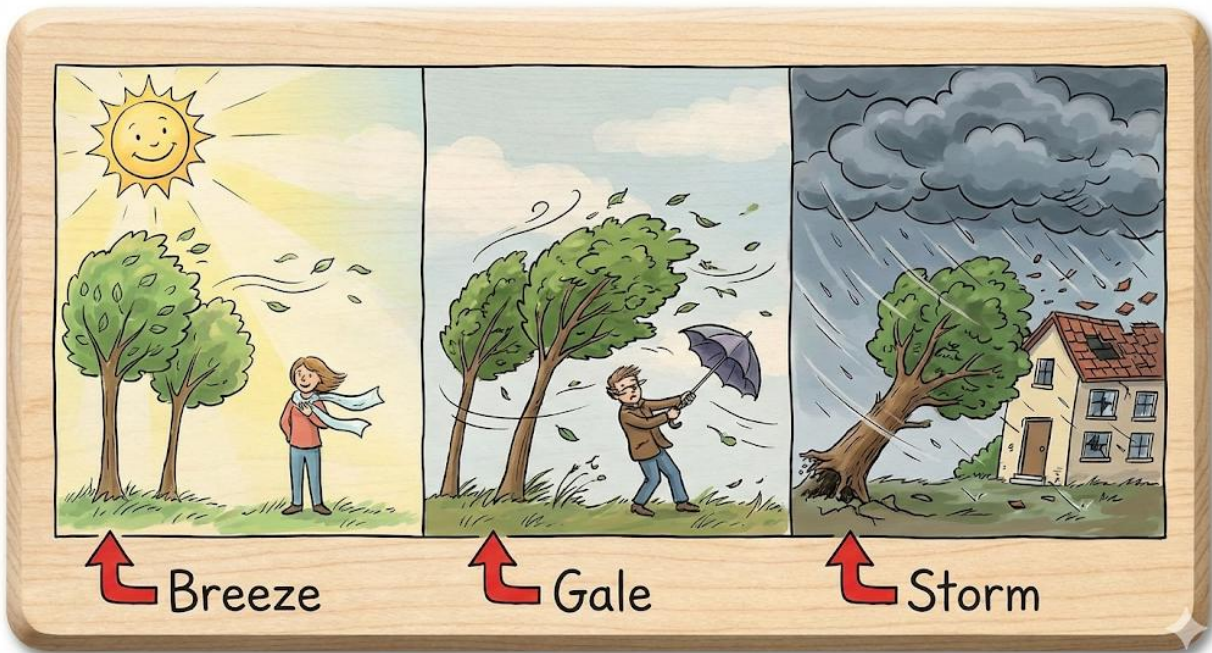
## CONTENT

**What is Wind?** Wind is simply **moving air**.

- When air moves from one place to another, we feel it as wind.
- Air usually moves from a place of **high pressure** (where air is squeezed together) to a place of **low pressure** (where there is more space).

**Types of Wind:** Winds can be gentle or very strong. We give them different names:

- **Breeze:** A **gentle, soft, and pleasant** wind. It makes leaves rustle softly and feels cool on a hot day. It is safe and helpful.
- **Gale/Strong Wind:** A **fast and powerful** wind. It can make trees sway a lot, blow dust around, and make it hard to walk straight. It often comes with heavy rain.
- **Storm/Hurricane/Tornado:** These are **very violent and dangerous** winds. They can uproot big trees, blow off rooftops, destroy farms, and cause floods. We must stay indoors when there is a storm warning.



**Uses of Wind (Good Things Wind Does):** Wind is very useful when it is not too strong:

- **Drying:** It helps to dry our wet clothes on the line, dry crops like pepper and grains, and dry fish.
- **Cooling:** A cool breeze makes us feel better on a hot day.
- **Transportation:** It pushes sailing boats and ships across water. Long ago, all ships used wind.
- **Energy:** Wind turns the big blades of **windmills** or **wind turbines**. These machines can pump water or generate electricity (wind power).
- **Recreation:** Wind helps us to fly kites and is used in sports like windsurfing.
- **For Nature:** It helps to scatter seeds from plants and pollinate flowers.

**Dangers of Strong Wind:** Very strong wind can cause harm:

- It can **destroy houses**, blow off roofs, and break windows.
- It can **uproot trees** and flatten farms, destroying crops.
- It can cause **accidents** by blowing vehicles off course or causing falling branches.
- It can lead to **soil erosion** by blowing away the topsoil.



## EVALUATION

1. What is the simple definition of wind?
2. Which type of wind would be perfect for flying a kite on a sunny day?
3. Name two ways people use wind to help them in their work.
4. How can a very strong wind be dangerous to a farmer's crop?
5. What natural machine uses wind to make electricity?

## ASSIGNMENT

- **(Practical):** Make a simple paper pinwheel or a ribbon wand. Take it outside and see if the wind makes it move. Describe what happens.
- **(Observation):** Look at trees or flags outside for one minute. Are they moving? If yes, would you call it a breeze or a strong wind?
- **(Written):** List two things in your community that could be damaged by a very strong storm.

## **WEEK 5: MIDTERM EXAMINATION.**

**CONTENT:** Assessment on Weeks 1-4 (Soil & Air) **EVALUATION:** Written Test

**ASSIGNMENT:** Review corrections.

---



**WEEK 6: MIDTERM BREAK.**

---





## WEEK 7: QUALITY OF WATER I (SOURCES).

**OBJECTIVES** By the end of the lesson, pupils should be able to:

- List four sources of water.
- State three domestic uses of water.
- Explain why water is important to life.

### CONTENT

**Sources of Water:** Where does the water we use come from? We get water from different places called sources.

- **Rain:** Water that falls from clouds in the sky. It is a natural and pure source, but it can pick up dirt from the air and roofs.
- **Rivers and Streams:** Natural flowing water on land. Many communities get water from rivers, but it often needs to be cleaned.
- **Lakes and Ponds:** Large bodies of still, fresh water collected in hollows on land.
- **Wells:** Deep holes dug into the ground to reach water that has soaked into the soil (groundwater). A bucket is used to draw the water.
- **Boreholes:** Deep, narrow holes drilled by a machine to reach groundwater. A pump brings the water up through a pipe, often to a tap.
- **Tap Water:** This water usually comes from a treatment plant that cleans water from rivers or boreholes and pumps it to our homes through pipes.
- **Springs:** Places where groundwater naturally flows out of the ground.
- **Sea/Ocean:** This is a huge source, but the water is **salty** and cannot be used for drinking or farming unless the salt is removed (which is very expensive).



**Importance of Water (Why We Need It):** Water is the most important thing for all living things. We cannot live without water.

- **For Our Bodies:** We need to drink water every day to stay alive. Our bodies are mostly made of water! It helps us digest food, control body temperature, and remove waste.
- **For Plants:** Plants need water to grow, to make their food, and to stay firm. A plant without water wilts and dies.
- **For Animals:** All animals drink water.
- **For Cleaning:** We use water to wash our bodies, clothes, plates, and our homes to stay healthy and prevent diseases.
- **For Cooking:** We need water to prepare almost all our foods like rice, stew, tea, and soup.

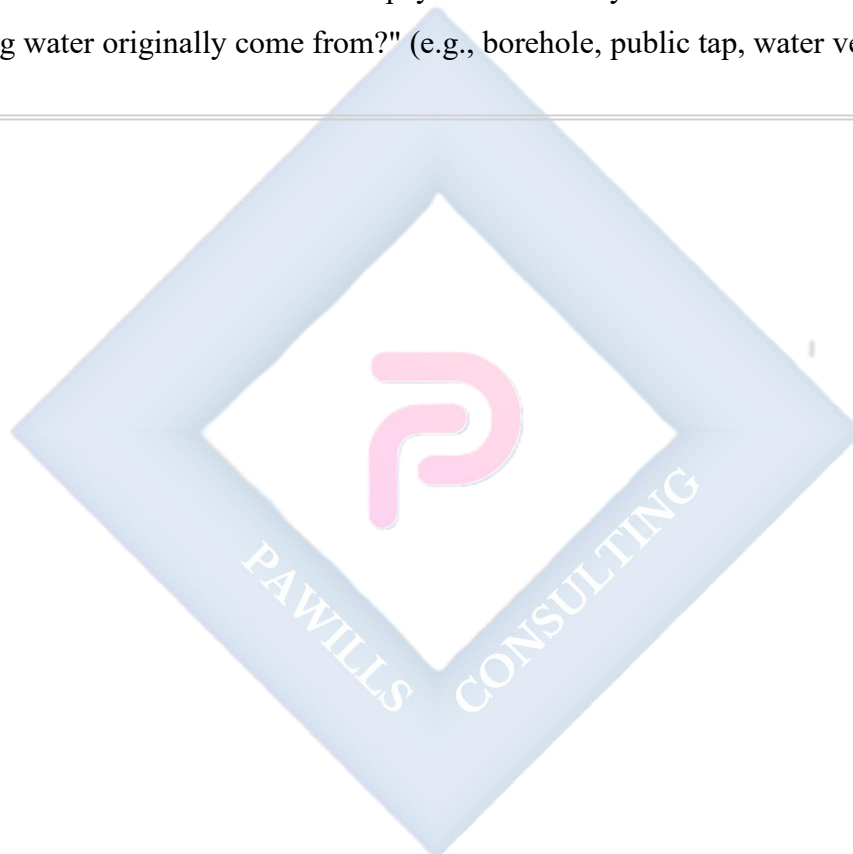
## EVALUATION

1. Where does rain come from?
2. What is the difference between a well and a borehole?
3. Why can't we drink water directly from the ocean?
4. Give two reasons why your body needs water.

5. What would happen to a potted plant if you stopped watering it?

### ASSIGNMENT

- **(Drawing):** Draw a source of water that your family uses (e.g., a tap, a well, a water tank).
  - **(Practical/Written):** From the time you wake up until you go to bed, list four different times you used water today.
  - **(Oral):** Ask the adult who fetches or pays for water in your house: "Where does our drinking water originally come from?" (e.g., borehole, public tap, water vendor).
- 



## WEEK 8: QUALITY OF WATER II (CLEAN VS DIRTY).

**OBJECTIVES** By the end of the lesson, pupils should be able to:

- Differentiate between clean and dirty water.
- List three qualities of clean (potable) water.
- Mention two diseases caused by drinking dirty water.
- Identify ways to make water clean.

### CONTENT

**Clean Water vs. Dirty Water:** Not all water is safe to drink. We must be able to tell the difference.

- **Clean/Safe Water (Potable Water):** Water that is free from germs, dirt, and harmful chemicals. It is safe for drinking, cooking, and bathing.
- **Dirty/Unsafe Water:** Water that contains harmful things like mud, animal waste, rubbish, germs (bacteria and viruses), or chemicals. Drinking it can make us very sick.

**Qualities of Clean Water:** How can we know water is clean? We use our senses:

- **Clear/Colourless:** Clean water has no colour. You can see through it clearly. It is not brown, green, or muddy.
- **Odourless:** Clean water has no smell. It should not smell like eggs, chemicals, or anything rotten.
- **Tasteless:** Clean water has no strong taste. It should not taste salty, sour, or bitter.
- **Free from Particles:** You should not see any sand, leaves, worms, or rubbish floating in it.



**Dangers of Dirty Water:** Drinking or using dirty water causes many serious sicknesses. These are called **water-borne diseases**.

- **Cholera:** Causes severe vomiting and watery stooling, leading to quick dehydration. It can be deadly.
- **Typhoid Fever:** Causes high fever, stomach pain, headache, and weakness.
- **Dysentery:** Causes stomach pain and bloody stool.
- **Diarrhea:** Very common in children; causes the body to lose too much water.

**Making Water Safe for Drinking:** We can treat dirty water at home:

- **Boiling:** This is the best and easiest method. Heat the water until it bubbles and boils for at least 5 minutes. This kills almost all germs. Let it cool before drinking.
- **Filtering:** Pouring water through a clean cloth, fine sieve, or a water filter pot removes dirt and some germs.
- **Adding Water Guard (Chlorine):** Using special drops or tablets (like Water Guard) kills germs in the water.



- **Settling & Decanting:** Let muddy water sit in a bucket for some hours. The dirt will settle at the bottom. Then gently pour the clearer water from the top into another container.

## EVALUATION

1. Should clean water have a smell or a colour?
2. If you fetch water from a stream and it looks brown, is it safe to drink? Why?
3. Name one sickness you can get from drinking dirty water.
4. What is the simplest method to kill germs in water at home?
5. True or False: Clear water is always safe to drink. (Hint: Some germs are invisible!).

## ASSIGNMENT

- **(Practical Observation):** Take a clear glass cup. Fill it with drinking water from your home. Look at it against the light. Is it clear? Smell it. Write one sentence about what you observe.
  - **(Written):** Write down two important reasons why we should never drink water from a dirty pond or puddle.
  - **(Drawing):** Draw a poster with a simple message: "Always Drink Clean Water!"
-

## WEEK 9: USES OF WATER.

**OBJECTIVES** By the end of the lesson, pupils should be able to:

- Categorize the uses of water into Domestic, Agricultural, and Industrial.
- Mention two uses of water for recreation (fun).
- Explain the importance of water for transportation.
- State one way to conserve water.

## CONTENT

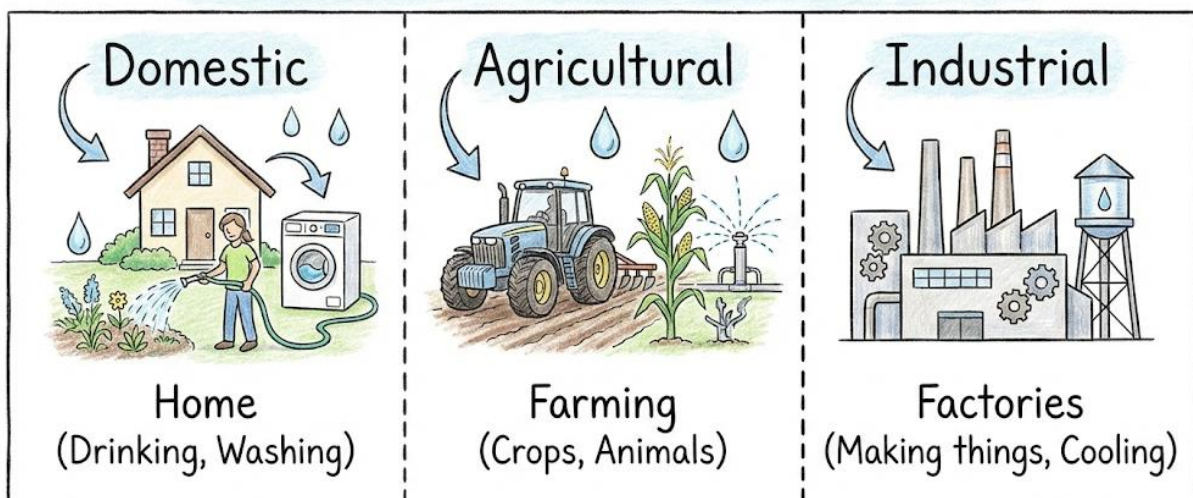
**Categories of Water Use:** People use water in many ways. We can group these uses into three main categories:

- **Domestic Use (Use at Home):** This is how we use water in our houses every day.
  - Drinking and Cooking
  - Bathing and Washing hands
  - Washing clothes and dishes
  - Cleaning the house and compound
  - Flushing toilets
- **Agricultural Use (Use in Farming):** This is how farmers use water to produce our food.
  - **Irrigation:** Watering crops on the farm when there is no rain. This ensures plants grow well all year.
  - **Watering Livestock:** Giving water to animals like cows, goats, chickens, and fish in ponds to drink.
  - **Processing Farm Produce:** Washing vegetables, fruits, and cassava after harvest.
- **Industrial Use (Use in Factories and Industries):** This is how big companies and factories use water.



- **Cooling Machines:** Water is used to cool hot machines in factories so they don't overheat.
- **Cleaning and Washing:** Used to wash raw materials and clean factory floors.
- **Mixing Ingredients:** Water is mixed with other things to make products like drinks, medicines, and paints.
- **Other Important Uses:**
  - **Transportation:** Ships, boats, canoes, and ferries carry people and goods across rivers, lakes, and oceans. Water provides a "road" for them.
  - **Recreation (For Fun and Sports):** Swimming, playing in the pool, diving, surfing, fishing, and boat riding.
  - **Generating Electricity (Hydropower):** Fast-flowing water in dams (like Kainji Dam) is used to turn turbines and produce electricity.

## How Do We Use Water?



**Conservation of Water (Saving Water):** Water is precious. We must not waste it because clean water is not easy to get for everyone. Here is how to save water:

- **Turn off the tap** tightly after use. A dripping tap wastes a lot of water.
- **Take shorter showers** instead of very long baths.
- **Collect rainwater** in a tank or bucket to water plants or wash things.
- **Use a bucket of water** to wash the car instead of a running hose.
- **Report leaking pipes** or taps to an adult immediately.

## EVALUATION

1. Is washing plates a domestic, agricultural, or industrial use of water?
2. Why is irrigation very important for farmers in the dry season?
3. Name one type of vehicle that uses water as its "road."
4. What fun activity can you do in a swimming pool?
5. Give one example of how you can save water at home.

## ASSIGNMENT

- **(Drawing):** Draw a boat carrying people on a river.
  - **(Written):** List three ways you or your school use water in one day (e.g., drinking, washing hands, watering flowers).
  - **(Practical):** Tonight, when brushing your teeth, turn off the tap while you are brushing. Tell your family why this saves water.
-

## WEEK 10: PLANTS I (PARTS AND FUNCTIONS).

**OBJECTIVES** By the end of the lesson, pupils should be able to:

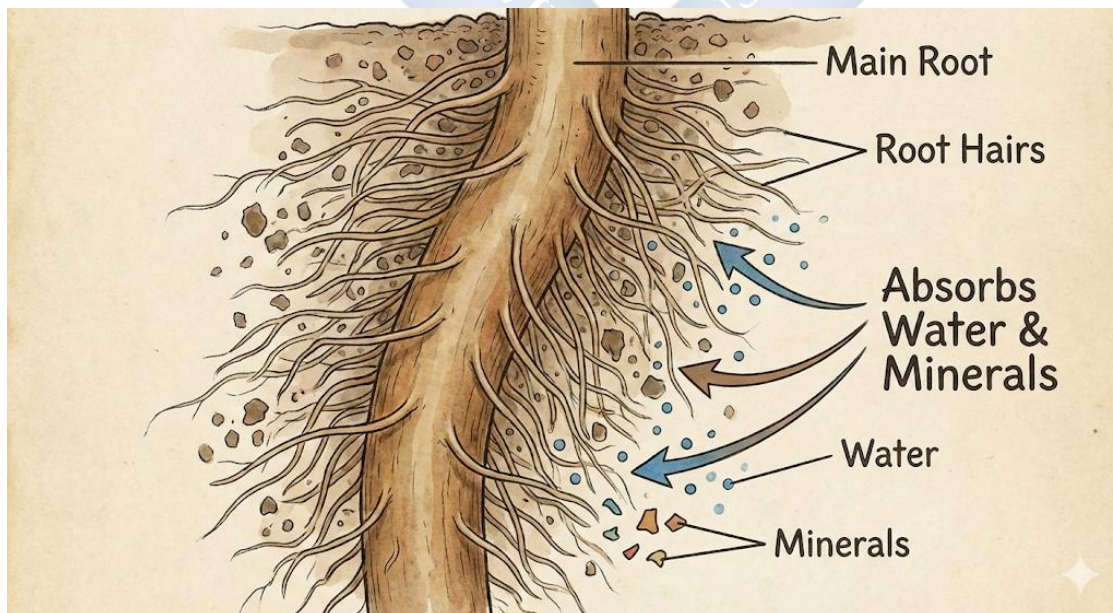
- Define plants as living things.
- Draw and label a plant showing the root, stem, leaf, and flower.
- State the function of the roots and leaves.

### CONTENT

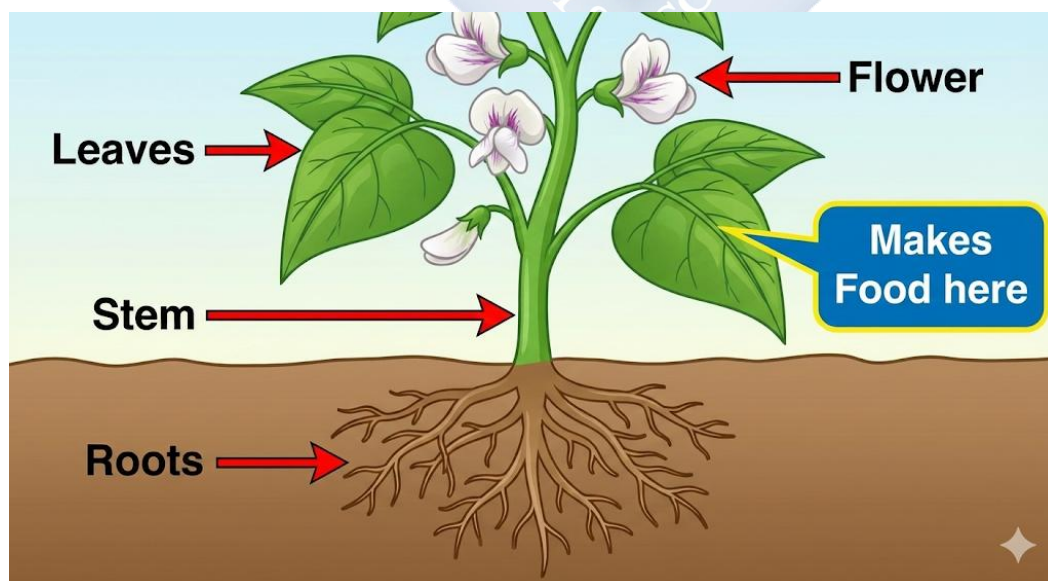
**What are Plants?** Plants are **living things**. They grow, they need food and water, they breathe, and they can have babies (reproduce). Unlike animals, most plants are fixed in one place by their roots. They make their own food using sunlight!

**Parts of a Plant and Their Jobs (Functions):** A plant is like a factory with different departments, each with a special job.

- **Roots:** Usually found underground.
- **Function 1: Anchor/Support.** They hold the plant firmly in the soil so it doesn't fall over when the wind blows.
- **Function 2: Absorption.** They have tiny hair-like parts that suck up water and minerals (plant food) from the soil.



- **Stem:** The main "body" or "trunk" of the plant. It stands above the ground.
- **Function 1: Support.** It holds up the leaves, flowers, and fruits towards the sunlight.
- **Function 2: Transportation.** It has tiny tubes (like pipes) that carry water and minerals from the roots up to the leaves. It also carries food made in the leaves down to other parts of the plant.
- **Leaves:** Usually flat, thin, and green.
- **Main Function: Photosynthesis.** This is a special process. Leaves use sunlight (energy), water (from roots), and carbon dioxide (from air) to cook up the plant's food (sugar/glucose). They release oxygen into the air as a waste product, which is what we breathe!
- **Other Function: Breathing (Respiration).** Plants also breathe through tiny holes on the underside of leaves called **stomata**.
- **Flowers:** The often colourful and beautiful part.
- **Main Function: Reproduction.** Flowers attract insects and birds with their colour and smell. These visitors help to pollinate the flower, which leads to the formation of **fruits and seeds**. Seeds grow into new plants.



## EVALUATION

1. Are plants living or non-living? Give one reason for your answer.
2. Which part of the plant is usually hidden under the soil?
3. What is the main job of the leaves? (Hint: It starts with 'Photo...')
4. If you pull a plant out of the soil, why does it die?
5. What do flowers eventually turn into?

## ASSIGNMENT

- **(Practical Observation):** With permission, carefully pull out a small weed (like grass) from the ground. Look closely at its roots. Wash your hands afterwards. Draw what you saw.
  - **(Drawing):** In your notebook, draw a complete plant and label these four parts: Root, Stem, Leaf, Flower.
  - **(Written):** Why is the work done by leaves important not just for the plant, but also for us humans and animals? (Think about what they release into the air).
-



**WEEK 11: REVISION.**

**CONTENT:** Review of all Term 2 topics (Soil, Air, Water, Plants)

**EVALUATION:** Oral & Written Questions

**ASSIGNMENT:** Study for exams.



## PRIMARY 3 BASIC SCIENCE SCHEME OF WORK

### THIRD TERM

| Week | Topic                      | Content                                                                                                                                                                    |
|------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Plants II                  | Types of plants (herbs, shrubs, trees); Examples of each type; Differences between them                                                                                    |
| 2    | Plants III                 | Uses of plants (food, medicine, shelter, clothing, furniture); Importance of plants; Taking care of plants                                                                 |
| 3    | Animals I                  | What are animals? Characteristics of animals (they move, eat, grow, reproduce); Differences between plants and animals                                                     |
| 4    | Animals II                 | Types of animals (domestic and wild); Examples of domestic animals (dog, cat, goat, cow, chicken); Examples of wild animals (lion, elephant, monkey)                       |
| 5    | <b>Midterm Examination</b> | Assessment of weeks 1-4 topics                                                                                                                                             |
| 6    | <b>Midterm Break</b>       |                                                                                                                                                                            |
| 7    | Animals III                | Uses of animals (food, work, companionship, clothing); Taking care of animals; Kindness to animals                                                                         |
| 8    | Forms of Technology I      | Meaning of technology; Technology in our homes (television, radio, telephone, computer, fan, iron); How technology makes life easier                                       |
| 9    | Forms of Technology II     | Technology in schools (computers, projectors, calculators); Technology in hospitals (X-ray machines, thermometers); Technology in transportation (cars, trains, airplanes) |
| 10   | Forms of Energy - Light I  | What is light? Sources of light (sun, moon, fire, bulb, candle, torch); Natural and artificial sources of light; Importance of light                                       |
| 11   | <b>Revision</b>            | Review of all topics covered in third term and entire year                                                                                                                 |
| 12   | <b>Examination</b>         | End of year assessment                                                                                                                                                     |
| 13   | <b>Closing</b>             | Collection of results; Distribution of report cards; Closing ceremony                                                                                                      |



## PRIMARY 3 BASIC SCIENCE LESSON NOTES

### THIRD TERM

#### WEEK 1: PLANTS II (TYPES OF PLANTS).

**OBJECTIVES** By the end of the lesson, pupils should be able to:

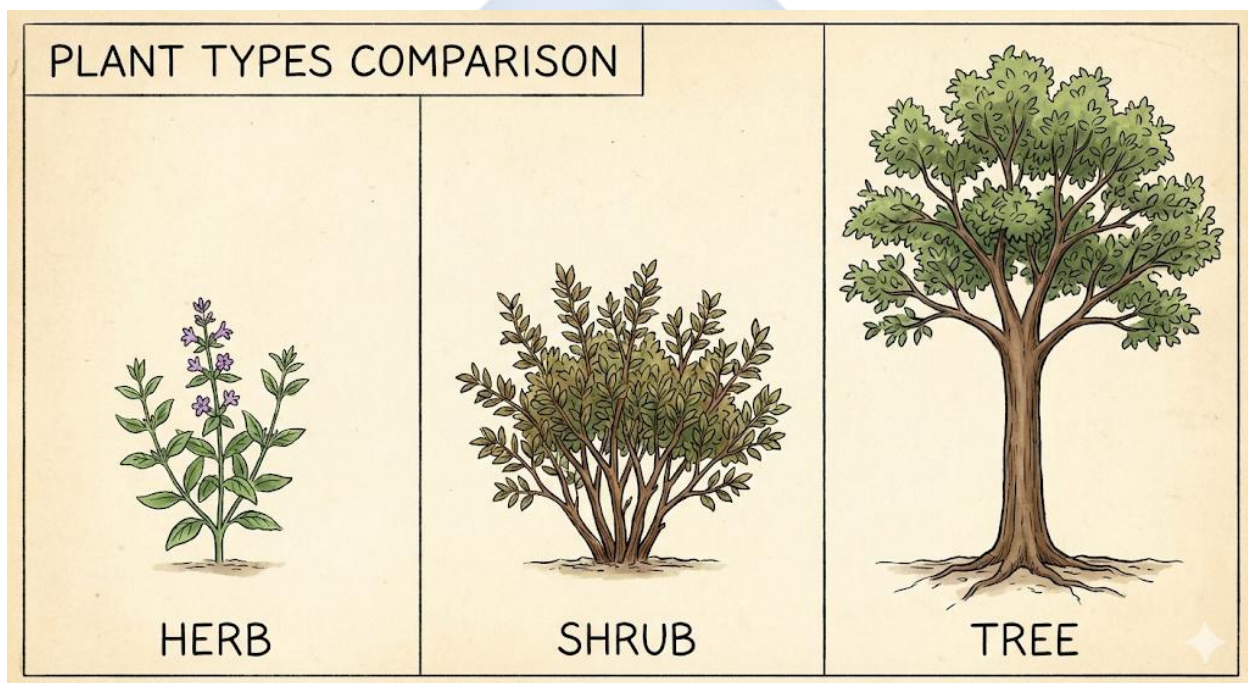
- List the three main types of plants based on size and stem texture.
- Define Herbs, Shrubs, and Trees.
- Give two examples for each type of plant.
- State one difference between a shrub and a tree.

#### CONTENT

**Types of Plants:** Plants come in all shapes and sizes. We can group them into three main types based on how tall they grow and what their stems feel like.

- **Herbs:**
- **Description:** These are **small, soft plants**. They have **green, soft, and juicy stems** that can be easily bent or broken. They are usually short and do not live for many years. Many herbs are plants we eat.
- **Examples:** Tomato plant, Pepper plant, Spinach (Efo tete), Waterleaf, Grass, Maize plant, Lettuce.
- **Key Feature:** Soft, green stem. Often finish their life cycle in one season.
- **Shrubs:**
- **Description:** These are **medium-sized plants**, bigger than herbs but smaller than trees. They have **hard, woody stems** that are not very thick. Shrubs have many branches starting close to the ground, so they look like thick bushes. They live for several years.
- **Examples:** Hibiscus flower (Common zobo plant), Rose plant, Sunflower, Cocoyam plant, Cassava plant, Cotton plant.

- **Key Feature:** Many woody stems from the base. Bushy appearance.
- **Trees:**
- **Description:** These are **big, tall, and strong plants**. They have one main, thick, hard, woody stem called a **trunk**. The trunk supports heavy branches high above the ground. Trees live for many, many years—some for hundreds of years!
- **Examples:** Mango tree, Orange tree, Coconut tree, Iroko tree, Neem (Dogoyaro) tree, Palm tree, Mahogany tree.
- **Key Feature:** Single, tall, thick trunk.



### Simple Differences

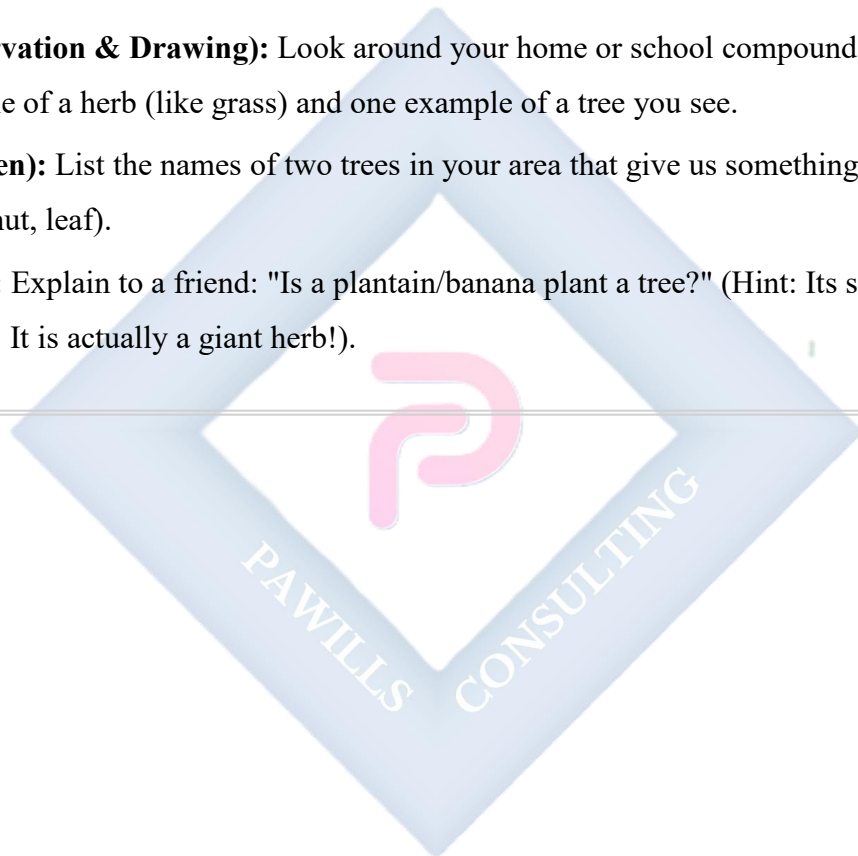
- **Herbs vs. Shrubs/Trees:** Herbs have soft stems you can pinch; shrubs and trees have hard, woody stems.
- **Shrubs vs. Trees:** Shrubs are shorter and have many stems from the ground; trees are taller with one main trunk before the branches start.

### EVALUATION

1. What are the three main groups of plants?
2. Which type of plant has a stem that is green and easy to bend?
3. Is a cassava plant a herb, shrub, or tree? (Hint: It has a hard stem but is not very tall with one trunk).
4. Give two examples of plants that are trees.
5. What is the special name for the big, hard stem of a tree?

## ASSIGNMENT

- **(Observation & Drawing):** Look around your home or school compound. Draw one example of a herb (like grass) and one example of a tree you see.
- **(Written):** List the names of two trees in your area that give us something useful to eat (fruit, nut, leaf).
- **(Oral):** Explain to a friend: "Is a plantain/banana plant a tree?" (Hint: Its stem is soft, not woody. It is actually a giant herb!).



## WEEK 2: PLANTS III (USES AND CARE).

**OBJECTIVES** By the end of the lesson, pupils should be able to:

- List four uses of plants to human beings.
- Mention two plants used for medicine.
- Explain why plants are important for the air we breathe.
- State two ways to take care of plants.

## CONTENT

**Uses of Plants:** We depend on plants for almost everything in our lives. They are super important!

- **Food:** Plants give us most of our food.
- **Fruits:** Mango, Orange, Pawpaw, Pineapple.
- **Vegetables:** Spinach, Cabbage, Tomato, Onions.
- **Grains & Seeds:** Rice, Beans, Maize, Wheat (for bread).
- **Roots & Tubers:** Yam, Cassava, Potato, Carrot.
- **Spices:** Pepper, Ginger, Garlic, Curry leaves.
- **Medicine (Herbal Medicine):** Many plants have healing powers.
- **Neem (Dogonyaro) leaves:** Used to treat malaria fever and skin infections.
- **Aloe Vera gel:** Soothes burns and skin cuts.
- **Moringa (Zogale) leaves:** Very nutritious, used to boost health.
- **Lemon grass & Ginger:** Used for tea to treat cough and cold.
- **Shelter and Furniture:** Wood from trees is used to build houses, make roofs, doors, windows, and furniture like tables, chairs, and beds.

- **Clothing:** Fibers from plants are used to make clothes.
- **Cotton plant:** Gives cotton wool which is spun into thread to make fabrics like Ankara, shirts, and towels.
- **Jute plant:** Used to make sacks and ropes.
- **Fresh Air:** Plants are like Earth's lungs. They take in a gas we don't need (**carbon dioxide**) and give out a gas we must breathe to live (**oxygen**). This process is called **photosynthesis**.
- **Other Uses:** Paper (from wood pulp), rubber (from rubber tree), perfumes (from flowers), cooking oil (from palm, groundnut), and shade/beauty.



**Taking Care of Plants:** Since plants do so much for us, we must take good care of them.

- **Watering:** Give them water regularly, especially in the dry season. Don't flood them.
- **Weeding:** Remove unwanted plants (weeds) growing around them. Weeds steal water and nutrients.



- **Adding Manure/Fertilizer:** This is like plant food. It adds nutrients to the soil to help plants grow strong and healthy. Animal waste (like chicken droppings) can be used as manure.
- **Providing Sunlight:** Ensure they get enough sunlight. Don't plant a sun-loving plant in a very dark corner.
- **Protecting Them:** Do not break their stems or pluck leaves unnecessarily. Don't let animals trample on them.

## EVALUATION

1. Name three different kinds of food we get from plants.
2. Which plant is grown specially to make cloth for our clothes?
3. Name one piece of furniture at home that is made from wood.
4. Why is it a bad idea to cut down all the trees in a forest? (Think about air and animals).
5. If you have a flower in a pot, what are two things you must do to keep it alive?

## ASSIGNMENT

- **(Practical/Written):** Think about the meals you ate yesterday. List three foods that came directly from plants.
  - **(Oral):** Ask an elder in your family: "Which leaf or plant is used in our home to treat a common sickness like stomach ache or fever?" Write down the name of the plant.
  - **(Drawing):** Draw something useful in your house that is made from wood.
-

### WEEK 3: ANIMALS I (CHARACTERISTICS).

**OBJECTIVES** By the end of the lesson, pupils should be able to:

- Define animals as living things.
- List four characteristics of animals.
- State two differences between plants and animals.
- Identify animals in their immediate environment.

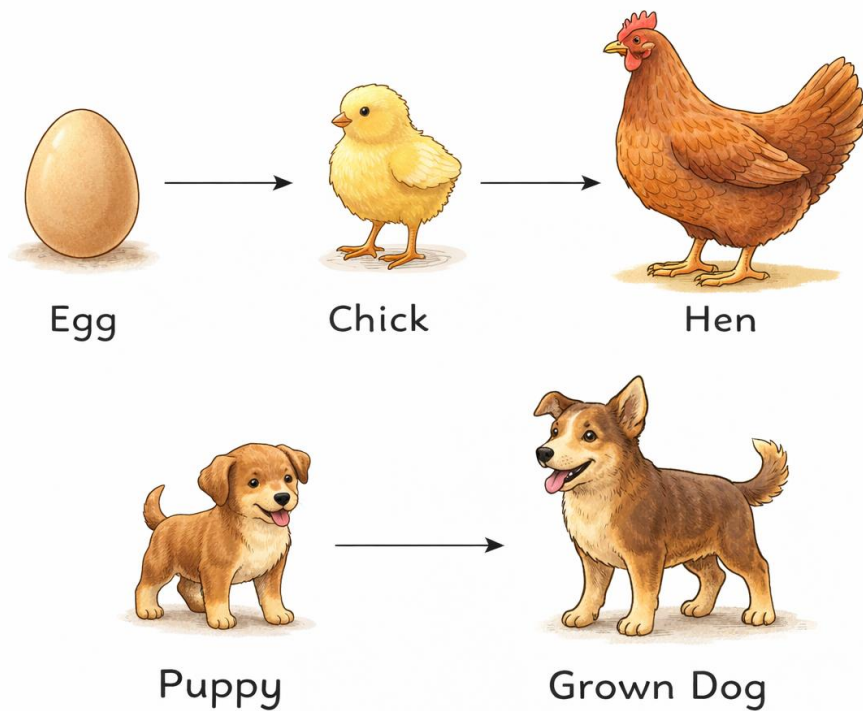
### CONTENT

**What are Animals?** Animals are **living things**. Like plants, they grow, breathe, need food, and can have young ones. But animals are very different from plants in special ways. The biggest difference is that **animals can move from place to place on their own** to find what they need.

**Characteristics of Animals (What All Animals Do):** We can remember what animals do with a simple code: **MRS GREN** (Movement, Respiration, Sensitivity, Growth, Reproduction, Excretion, Nutrition). For Primary 3, let's focus on these key ones:

- **Movement:** Animals move. They walk (goat), run (dog), fly (bird), swim (fish), crawl (snail), or jump (grasshopper). They move to find food, water, escape danger, or find a mate.
- **Feeding:** Animals **cannot make their own food** like plants. They must **eat** other living things (or things that were once living) to get energy. Some eat plants (goat - herbivore), some eat other animals (lion - carnivore), and some eat both (hen - omnivore).
- **Growth:** Animals start life small (as a baby, chick, puppy, kitten) and grow bigger and bigger until they become adults.
- **Reproduction:** Animals have babies to make more of their own kind. Some give birth to live young (cat → kittens), while others lay eggs that hatch later (hen → chicks).





- **Breathing (Respiration):** All animals breathe. Land animals use lungs to breathe air (oxygen). Fish use gills to breathe oxygen dissolved in water.
- **Sensitivity/Irritability:** Animals sense and react to their surroundings. They have sense organs (eyes, ears, nose, skin). If you touch a snail, it quickly goes into its shell. If you shine a light in a cockroach's path, it runs away.

**Differences Between Plants and Animals:** Let's compare:

| Feature             | Plants                                | Animals                                      |
|---------------------|---------------------------------------|----------------------------------------------|
| <b>Movement</b>     | Fixed in one place (roots in soil).   | Can move whole body from place to place.     |
| <b>Food</b>         | Make their own food (photosynthesis). | Must go and find/eat food.                   |
| <b>Sense Organs</b> | Do not have eyes, ears, or nose.      | Have sense organs to see, hear, smell, feel. |

| Feature       | Plants                               | Animals                                |
|---------------|--------------------------------------|----------------------------------------|
| <b>Growth</b> | Grow all their life, mainly at tips. | Grow to a certain adult size and stop. |

## EVALUATION

1. Are animals living things? Give one reason for your answer.
2. What is one thing **all** animals must do that plants do not need to do? (Hint: They go looking for it).
3. Can a plant run away from a goat that wants to eat it? Why?
4. How does a hen reproduce? (Does it give birth or lay eggs?)
5. What might a lizard do if you try to catch its tail? This shows which characteristic?

## ASSIGNMENT

- **(Observation):** Watch any animal around you (a bird, lizard, chicken, or even a housefly) for two minutes. Write down two actions it performed (e.g., "It flew to a branch," "It pecked the ground").
- **(Drawing):** Draw your favourite animal.
- **(Written):** Write one clear difference between a goat (animal) and a mango tree (plant).

## WEEK 4: ANIMALS II (TYPES OF ANIMALS).

**OBJECTIVES** By the end of the lesson, pupils should be able to:

- Classify animals into Domestic and Wild animals.
- Define domestic animals and give three examples.
- Define wild animals and give three examples.
- Identify where these animals live.

### CONTENT

**Types of Animals: Domestic vs. Wild:** We can group animals based on their relationship with humans.

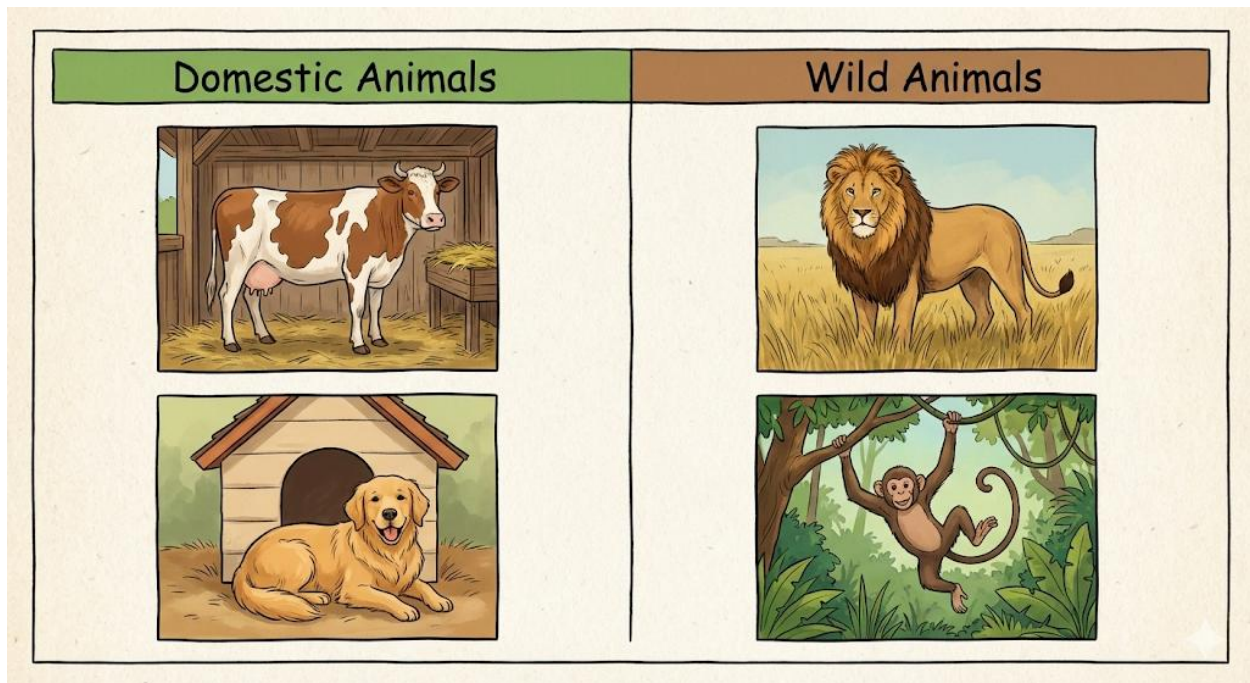
#### Domestic Animals:

- **Definition:** These are animals that **live with or near humans**. They have been **tamed** over many years. They are not usually afraid of people. We keep them for specific purposes like food, work, company, or protection.
- **Examples:**
  - **For Food/Products:** Cow, Goat, Sheep, Chicken, Duck.
  - **For Work/Transport:** Horse, Donkey, Bullock, Camel.
  - **For Companionship/Protection:** Dog, Cat, Rabbit.
  - **Where they live:** They live in homes or structures we build for them: **houses** (pets), **pens/sheds** (goats, cows), **coops** (chickens), **kennels** (dogs), or **stables** (horses).

#### Wild Animals:

- **Definition:** These are animals that **live freely in nature** (forest, bush, river, jungle). They are **not tamed** and are usually afraid of or dangerous to humans. They find their own food and shelter.
- **Examples:**

- **In Forests/Jungles:** Lion, Elephant, Monkey, Gorilla, Tiger.
- **In Rivers/Water:** Crocodile, Hippopotamus, Shark.
- **Others:** Snake, Hyena, Scorpion, Eagle, Rat (in the wild).
- **Where they live:** They live in natural homes: **dens** (lion), **holes/burrows** (rabbits in wild), **nests** (birds in trees), **caves, trees, or rivers**.



### Important Notes:

- Some animals can be both. For example, a rabbit can be a domestic pet or a wild animal. A cat can be a pet or a stray (wild) cat.
- We should **never try to touch or go near a wild animal** because they can be dangerous and might attack to protect themselves.

### EVALUATION

1. What does it mean for an animal to be "domestic"?
2. Give three examples of animals we keep at home or on the farm.
3. Where would you naturally find a wild animal like a hyena living?

4. Is a lion a domestic or wild animal? How do you know?
5. Why can't we keep a tiger or an elephant as a pet in our house like a dog?

### ASSIGNMENT

- **(Written):** Look around your neighbourhood. Make a list of five animals you have seen. Write "Domestic" or "Wild" next to each one.
  - **(Drawing):** Draw a picture of a domestic animal that gives us food.
  - **(Oral):** Explain to your parents why a squirrel in the tree outside is a wild animal, even though it comes close to houses.
- 



## **WEEK 5: MIDTERM EXAMINATION.**

**CONTENT:** Assessment on Weeks 1-4 (Plants & Animals) **EVALUATION:** Written Test

**ASSIGNMENT:** Review corrections.

---



**WEEK 6: MIDTERM BREAK.**

---





## WEEK 7: ANIMALS III (USES AND CARE).

**OBJECTIVES** By the end of the lesson, pupils should be able to:

- State four uses of animals to humans.
- Mention two animals used for work.
- Identify animals that give us food (meat, milk, eggs).
- List three ways to take care of animals.

## CONTENT

**Uses of Animals to Humans:** Animals help us in countless ways. We should be thankful for them.

- **Food:**
- **Meat:** We get meat from animals like Cow (beef), Goat (chevon), Chicken (poultry), Pork (pig), and Fish.
- **Milk:** Cows and Goats give us milk, which is used to make yogurt, cheese, and butter.
- **Eggs:** Hens (Chickens) and Ducks give us eggs, which are a great source of protein.
- **Work and Transportation:**
- **Carrying Loads:** Donkeys, Horses, Camels, and Bullocks (bulls) are used to carry heavy goods and people, especially in rural areas or deserts.
- **Ploughing Farms:** Bullocks are used to pull ploughs to prepare farmland for planting.
- **Guard and Security:** Dogs guard our homes, farms, and properties. They bark to alert us of strangers.
- **Companionship (Pets):** Animals like Dogs and Cats provide friendship and love. They play with us and make us happy, reducing loneliness.
- **Clothing and Materials:**

- **Wool:** Sheep give us wool, which is sheared, spun, and woven to make warm sweaters, socks, and carpets.
- **Leather:** The skin of cows, goats, and snakes is treated (tanned) to make leather for shoes, bags, belts, and jackets.
- **Silk:** Silkworms spin silk threads, which are used to make very soft and expensive cloth.
- **Other Uses:**
- **Manure:** The waste (dung) from cows, goats, and chickens is used as organic fertilizer to make soil fertile for plants.
- **Medicine:** Some animal parts are used in traditional medicine.

**Taking Care of Animals (Animal Welfare):** If we use animals, we have a responsibility to treat them well. This is called being **humane**.

- **Provide Good Food and Clean Water:** Animals need proper nutrition and fresh water daily.
- **Give Them Proper Shelter:** Their living place (kennel, coop, shed) should be clean, dry, and protect them from sun, rain, and extreme cold.
- **Healthcare:** When they are sick, they should be taken to a special animal doctor called a **Veterinary Doctor** or **Vet**.
- **Show Kindness:** Do not beat, kick, or throw stones at animals. They can feel pain and fear just like we do.

## EVALUATION

1. Name two animals that provide us with meat.
2. What do we get from a hen besides meat?
3. Which domestic animal is known for guarding the house?
4. What animal is commonly used in villages to pull a plough on a farm?
5. If you have a pet, what are two important things you must do for it every day?

## ASSIGNMENT

- **(Drawing):** Draw two sources of food from animals: a bottle of milk and an egg in a nest.
  - **(Written):** If you had a pet dog, what would you feed it? Where would it sleep?
  - **(Oral):** Ask around: "Who is a Veterinary Doctor and what do they do?" Prepare to tell the class.
- 



## WEEK 8: FORMS OF TECHNOLOGY I (IN THE HOME).

**OBJECTIVES** By the end of the lesson, pupils should be able to:

- Define technology simply as tools that make work easier.
- Identify five technological devices found in the home.
- State the function of each device.
- Explain how these devices make life easier.

### CONTENT

**Meaning of Technology:** Technology is **not just computers and phones**. It is any **tool, machine, or method** that humans create to solve a problem or make work easier, faster, or better. A simple spoon is technology—it helps us eat soup better than using our hands!

**Technology in Our Homes:** Our homes are filled with technology that helps with different tasks. Let's look at some common ones:

- **For Communication:**
- **Telephone/Mobile Phone:** Allows us to talk to and send messages to people who are far away instantly.
- **Radio/Television:** Brings news, music, educational programs, and entertainment from all over the world into our living room.
- **For Cooking and Food Preservation:**
- **Gas Cooker/Stove:** Provides instant, controllable heat for cooking food quickly and cleanly, unlike firewood.
- **Refrigerator/Freezer:** Keeps food fresh for days or weeks by making it cold. Prevents spoilage.
- **For Cleaning and Care:**
- **Electric Iron:** Uses heat to remove wrinkles from clothes, making them look neat.

- **Washing Machine:** Washes clothes automatically, saving time and energy compared to washing by hand.
- **Electric Blender:** Grinds and mixes food like tomatoes, peppers, and smoothies in seconds.
- **For Comfort:**
- **Electric Fan/Air Conditioner:** Cools the air in a room when the weather is hot.
- **Electric Heater/Warming Stove:** Warms a room when the weather is cold.

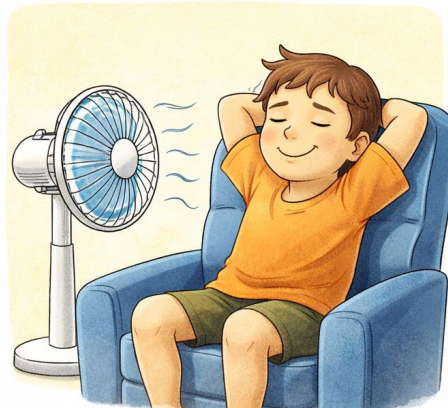
### How They Make Life Easier: A Simple Comparison

- **Without a fridge:** Food (like meat, milk) would spoil in one day. We'd have to go to the market every single day.
- **Without a phone:** To tell your grandma "Happy Birthday," you would have to travel to her village or send a letter that takes days.
- **Without a gas cooker:** You would have to gather firewood, make a fire, and get smoky every time you cook.

### Then and Now



Then



Now

## EVALUATION

1. Is a broom a form of technology? Why or why not?
2. Name three machines in your house that use electricity.
3. What is the main job of a refrigerator?
4. Which home technology helps us to see and hear events happening in other countries?
5. How did people press their clothes before the electric iron was invented? (Hint: They used a heavy metal tool heated on fire).

## ASSIGNMENT

- **(Drawing):** Draw two different technology items you have in your home.
- **(Oral Interview):** Ask your parents or grandparents: "What did you use to cool the house or cook food when you were young, before you had a fan or gas cooker?" Share the answer in class.
- **(Written):** Choose one home technology (e.g., a light bulb). Write one sentence on how life would be more difficult without it.

## WEEK 9: FORMS OF TECHNOLOGY II (SCHOOL, HOSPITAL, TRANSPORT).

**OBJECTIVES** By the end of the lesson, pupils should be able to:

- Identify technology used in schools.
- Identify technology used in hospitals.
- Identify technology used in transportation.
- State the use of a thermometer and an aeroplane.

### CONTENT

**Technology in Different Places:** Technology is used everywhere in our community to make services better.

#### 1. Technology in Schools:

- **Computer/Laptop:** Used for typing, researching on the internet, and learning with educational software.
- **Projector/Interactive Whiteboard:** Used by teachers to show pictures, videos, and notes to the whole class clearly.
- **Calculator:** Helps solve complex math problems quickly and accurately.
- **Public Address System (Microphone & Speakers):** Allows the headteacher to address the whole school during assembly.
- **Electric Bell:** Rings automatically to signal the start and end of lessons and break time.

**2. Technology in Hospitals/Clinics:** Hospitals use special technology to check our health and treat sickness.

- **Thermometer:** A small instrument used to measure body temperature. A high temperature means you have a fever.





- **Stethoscope:** Used by doctors and nurses to listen to sounds inside your body, like your heartbeat and breathing.
- **Syringe:** Used to give injections (vaccines, medicine) into the body.
- **X-ray Machine:** Takes a special picture of the inside of your body, like your bones, to see if they are broken.
- **Blood Pressure Monitor (Sphygmomanometer):** Checks the force of blood in your arteries.

**3. Technology in Transportation:** These are machines that help us move people and goods from one place to another.

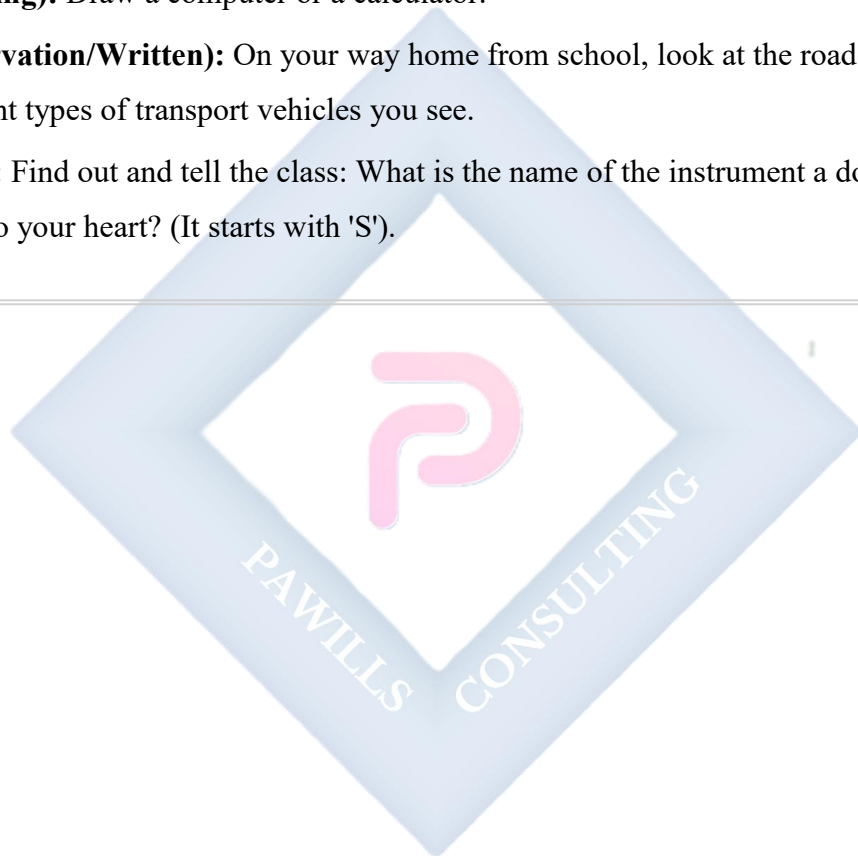
- **Land Transport:** Cars, Buses, Motorcycles (Okada), Bicycles, Trains.
- **Water Transport:** Boats, Canoes, Ships, Ferries.
- **Air Transport:** Aeroplanes, Helicopters. Aeroplanes can carry hundreds of people across countries and oceans in just a few hours.
- **Space Transport:** Rockets and Space Shuttles (the highest technology!), used to travel to space.

## EVALUATION

1. What technology does a teacher use to show a movie to the whole class?
2. Why does a nurse put a thermometer under your tongue?
3. Which form of transport is the fastest for travelling from Nigeria to England?
4. What do we use a calculator for in a mathematics lesson?
5. Name a vehicle that carries many people and moves on rails (tracks).

### ASSIGNMENT

- **(Drawing):** Draw a computer or a calculator.
- **(Observation/Written):** On your way home from school, look at the road. List three different types of transport vehicles you see.
- **(Oral):** Find out and tell the class: What is the name of the instrument a doctor uses to listen to your heart? (It starts with 'S').



## WEEK 10: FORMS OF ENERGY - LIGHT I.

**OBJECTIVES** By the end of the lesson, pupils should be able to:

- Define Light.
- List three natural sources of light.
- List three artificial (man-made) sources of light.
- State the importance of light.

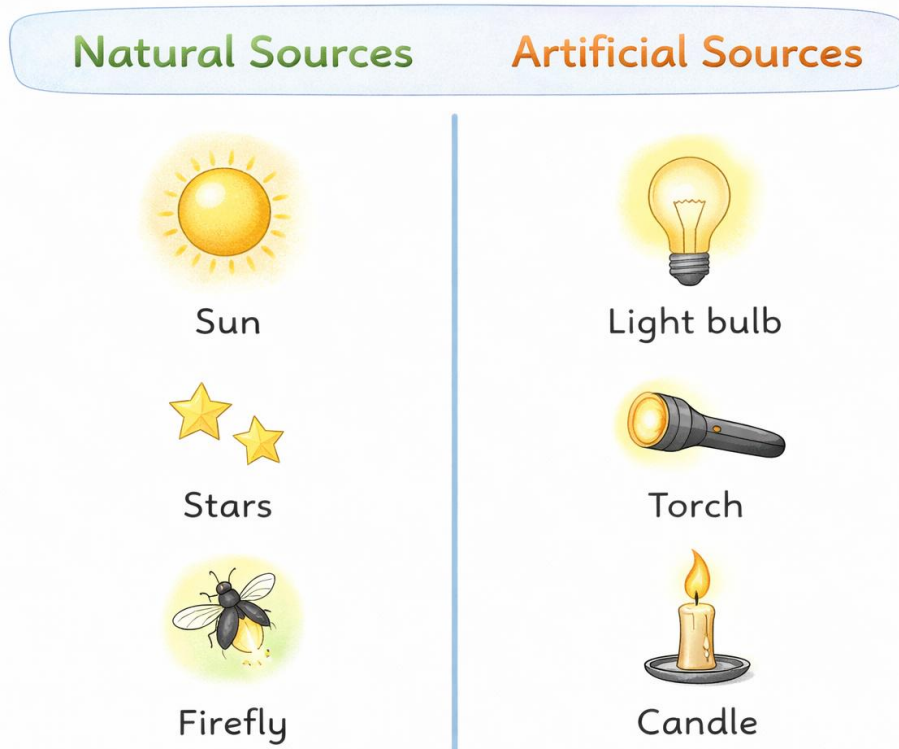
### CONTENT

**What is Light?** Light is a form of **energy** that our eyes can detect. It is what makes things **visible**. Without light, we would be in complete darkness and unable to see anything. Light usually travels in straight lines called **rays**.

**Sources of Light:** Anything that gives out its own light is called a **source of light**. We have two main types:

- **Natural Sources of Light:** These are made by nature.
- **The Sun:** This is the **most important** natural source. It gives light and heat to the whole Earth during the day.
- **The Moon & Stars:** The moon does not make its own light. It **reflects** light from the sun. Stars (like our sun, but very far away) make their own light and twinkle at night.
- **Lightning:** A huge spark of electricity during a thunderstorm that produces a flash of light.
- **Fireflies/Glow-worms:** Special insects that can produce light from their bodies (bioluminescence).
- **Artificial (Man-made) Sources of Light:** These are made by humans.
- **Electric Bulbs & Tubes:** Used in homes, schools, and streets.
- **Torchlight (Flashlight):** A portable light that uses batteries.

- **Candles & Lanterns (Kerosene or Battery):** Used when there is no electricity.
- **Car Headlights & Street Lights:** Help us see at night for safe travel.
- **Screen Light:** From televisions, phones, and computer monitors.



**Importance of Light:** Why do we need light?

- **For Vision (Seeing):** The most obvious reason. Light allows us to see the beautiful world, read, write, and play safely.
- **For Plants (Photosynthesis):** Plants need light (especially sunlight) to make their own food. No light, no plants. No plants, no food for us and animals.
- **For Warmth:** Sunlight provides natural warmth that dries clothes, keeps us warm, and helps in many daily activities.
- **For Safety:** Streetlights and traffic lights prevent accidents at night. Light helps us avoid tripping over things.

- **Regulates Our Body Clock:** Sunlight tells our body when to be awake (day) and when to sleep (night).

## EVALUATION

1. What is the main thing light helps us to do?
2. What is the biggest and most important natural source of light for Earth?
3. Name two artificial lights you can use when there is "NEPA" (power outage).
4. Why is light as important for plants as water is?
5. Does the moon produce its own light? If not, where does its light come from?

## ASSIGNMENT

- **(Drawing):** Draw the Sun during the day and a torch being used at night.
  - **(Practical):** Tonight, if there is a power outage, what will you use to find your way to the toilet? Write down the name of that light source.
  - **(Written):** Complete this sentence in your notebook: "Light is important because it helps us to \_\_\_\_\_ and helps plants to \_\_\_\_\_."
-

**WEEK 11: REVISION.**

**CONTENT:** Review of all Term 3 topics (Plants, Animals, Technology, Light)

**EVALUATION:** Oral & Written Questions

**ASSIGNMENT:** Study for end-of-year examination.

